

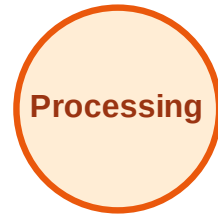
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

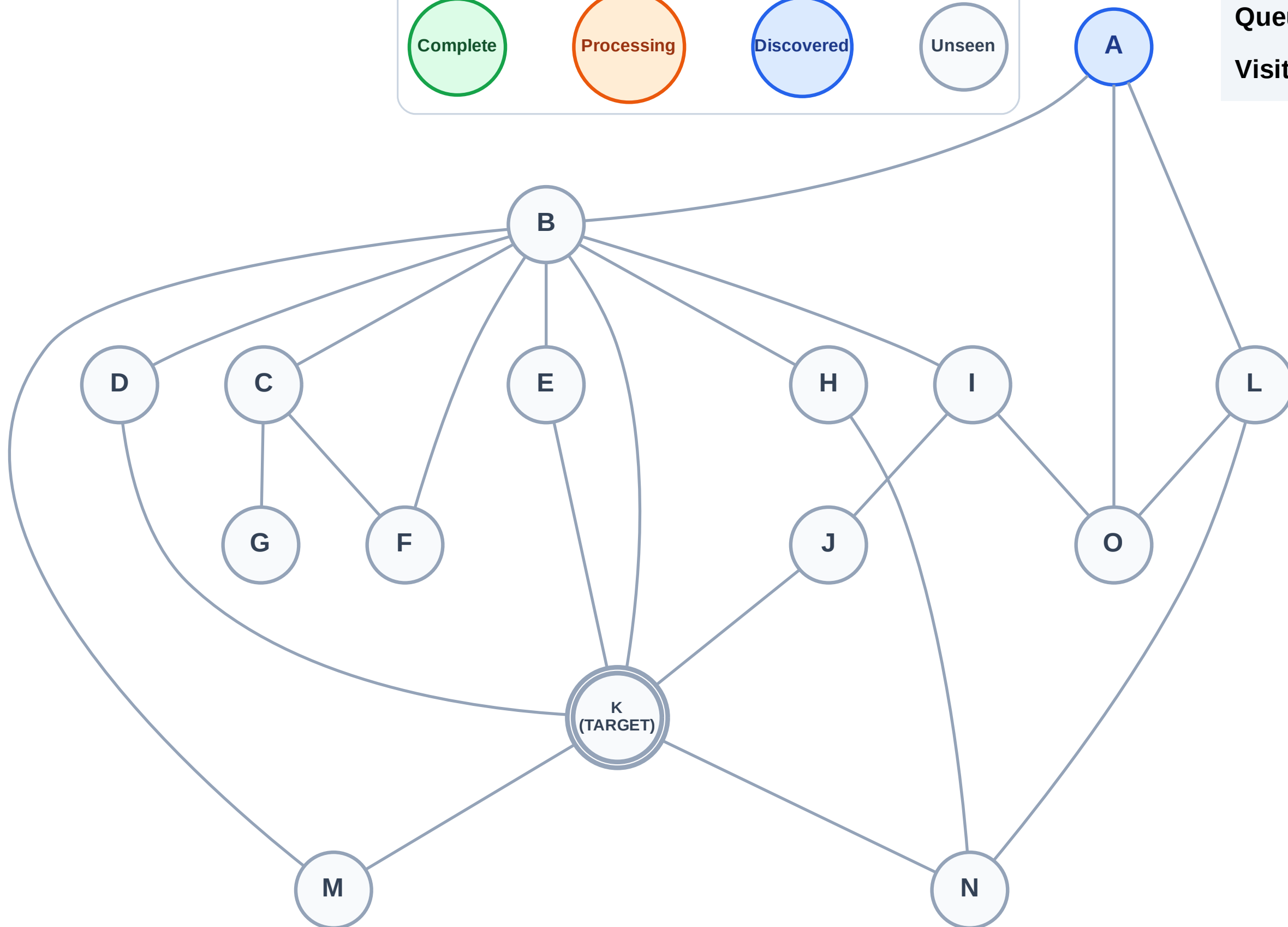


Discovered



Unseen

Queue: A
Visited: (none)



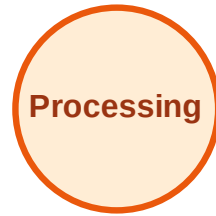
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing

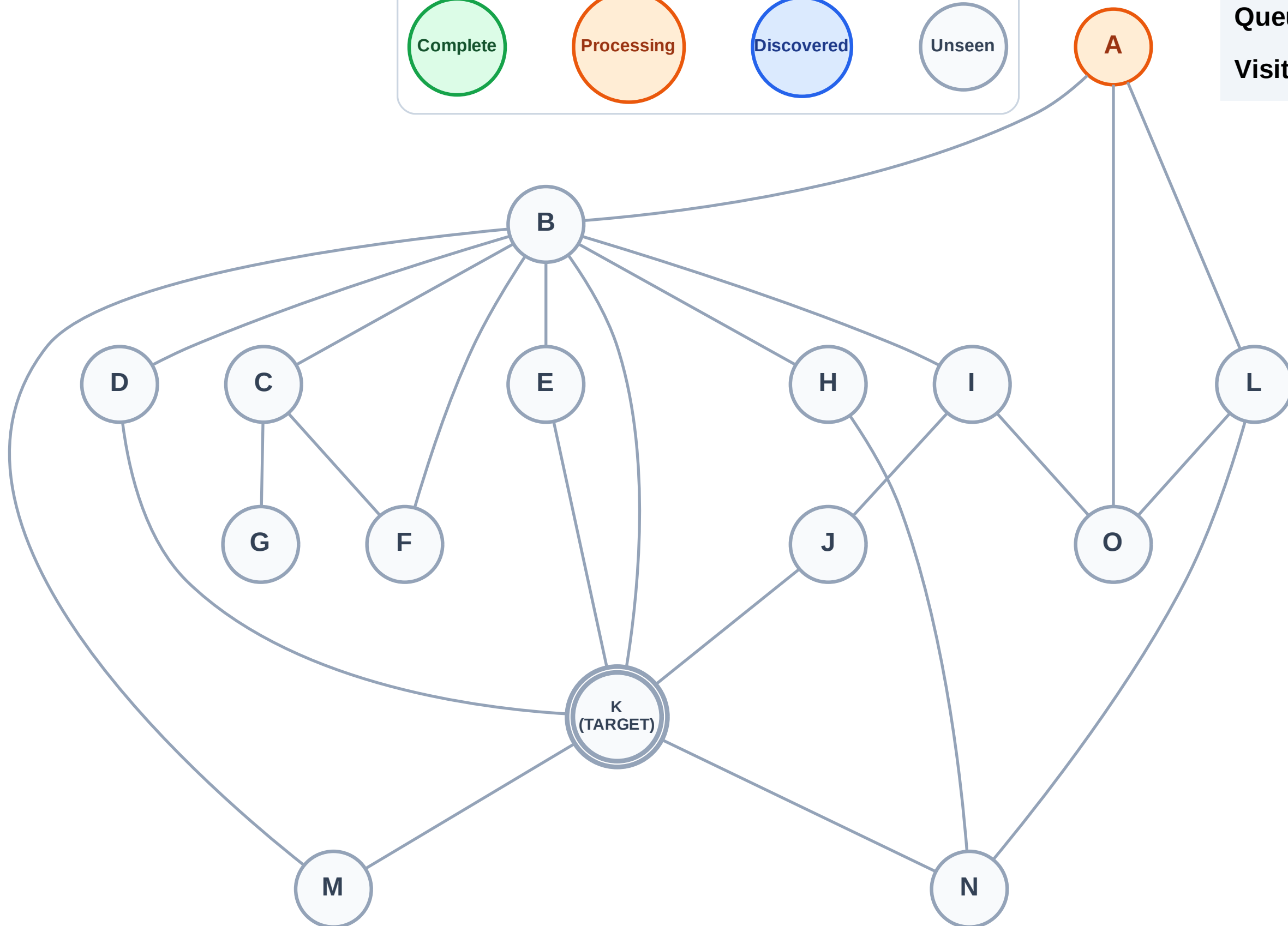


Discovered



Unseen

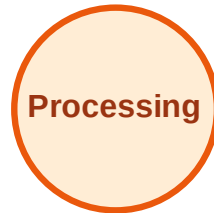
Queue: (empty)
Visited: (none)



BFS Traversal

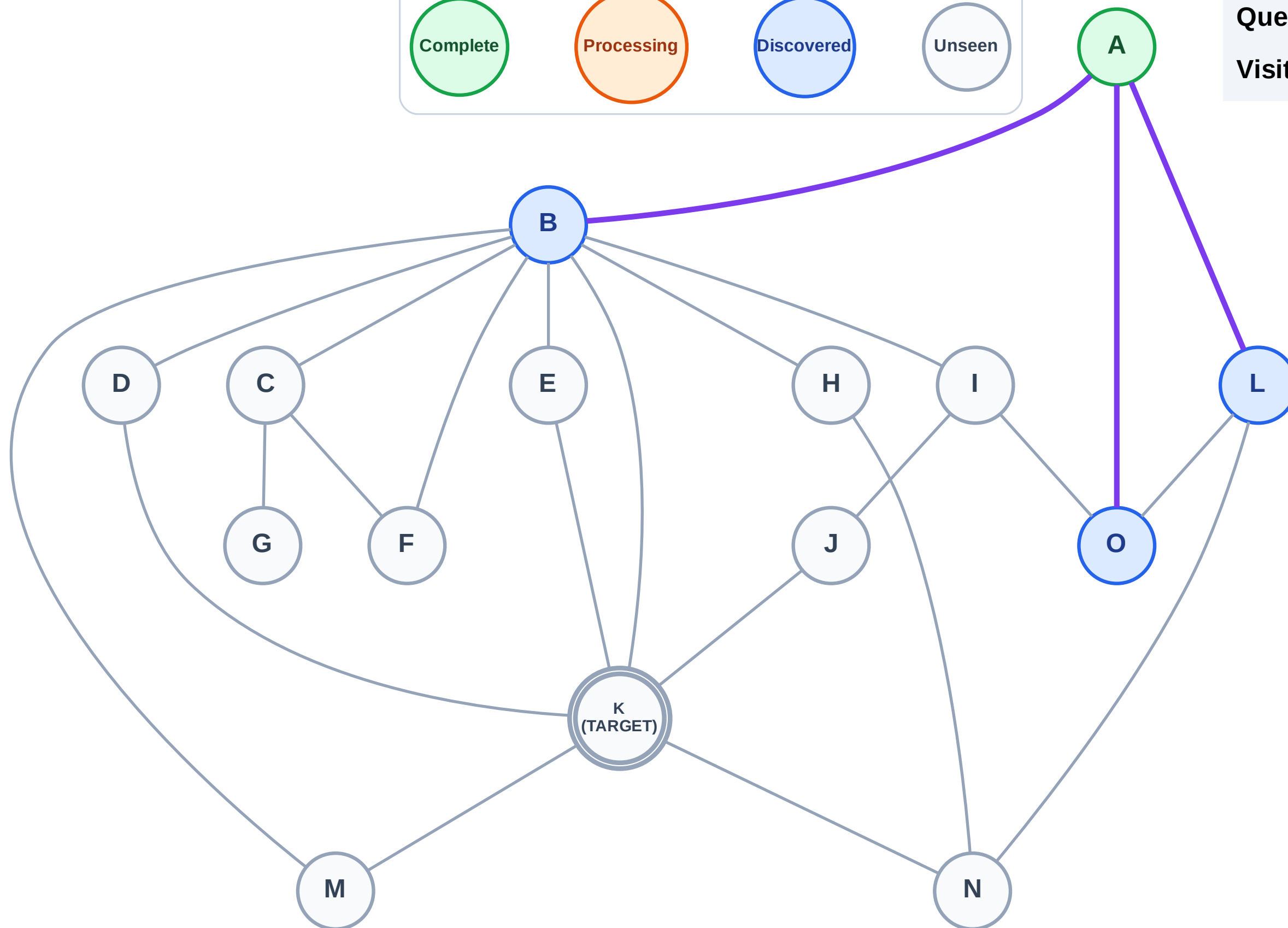
Step 3: Discover B, L, O from A

Legend



Queue: B → L → O

Visited: A



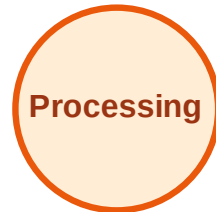
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



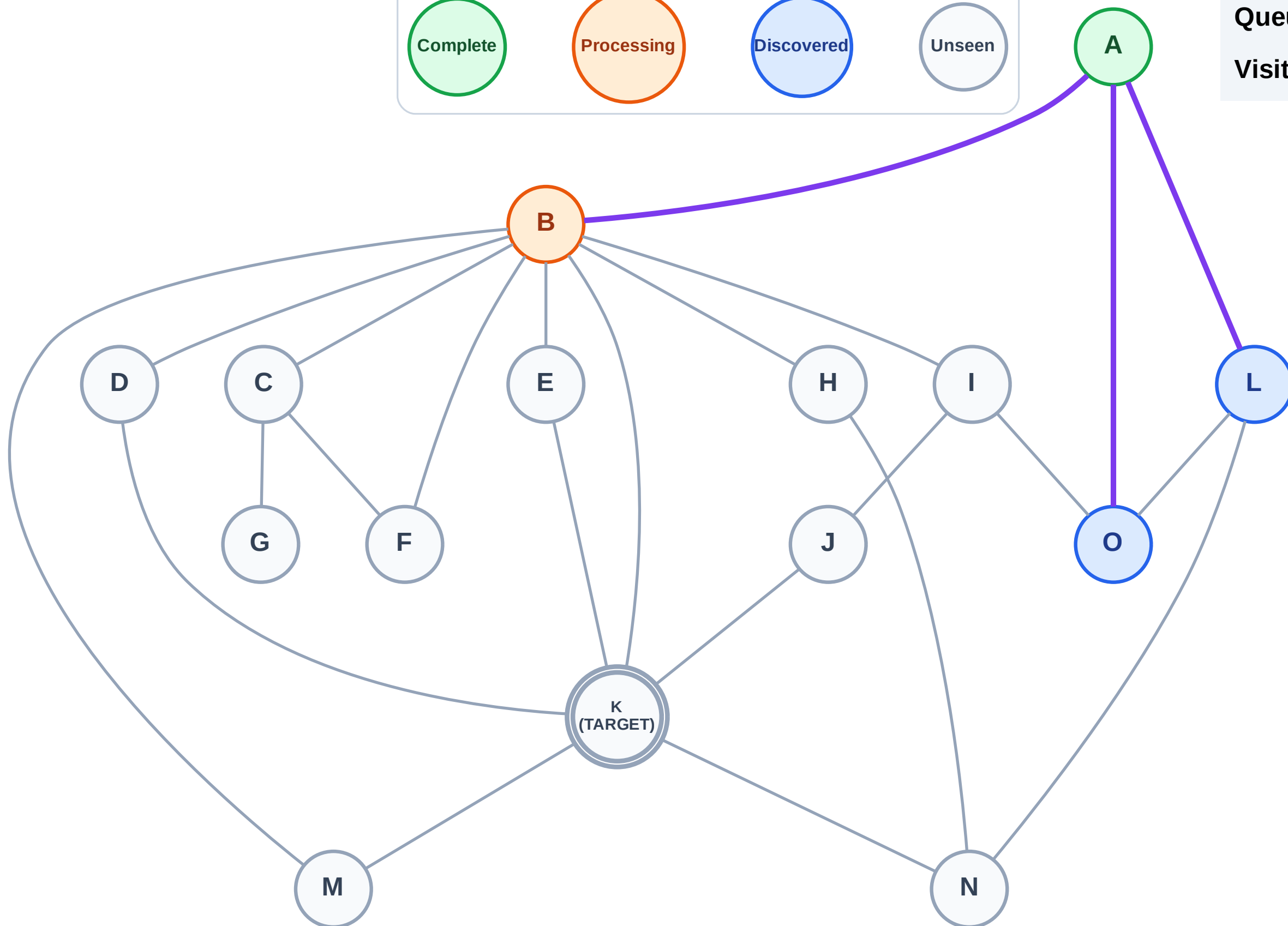
Discovered



Unseen

Queue: L → O

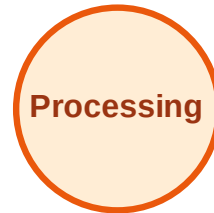
Visited: A



BFS Traversal

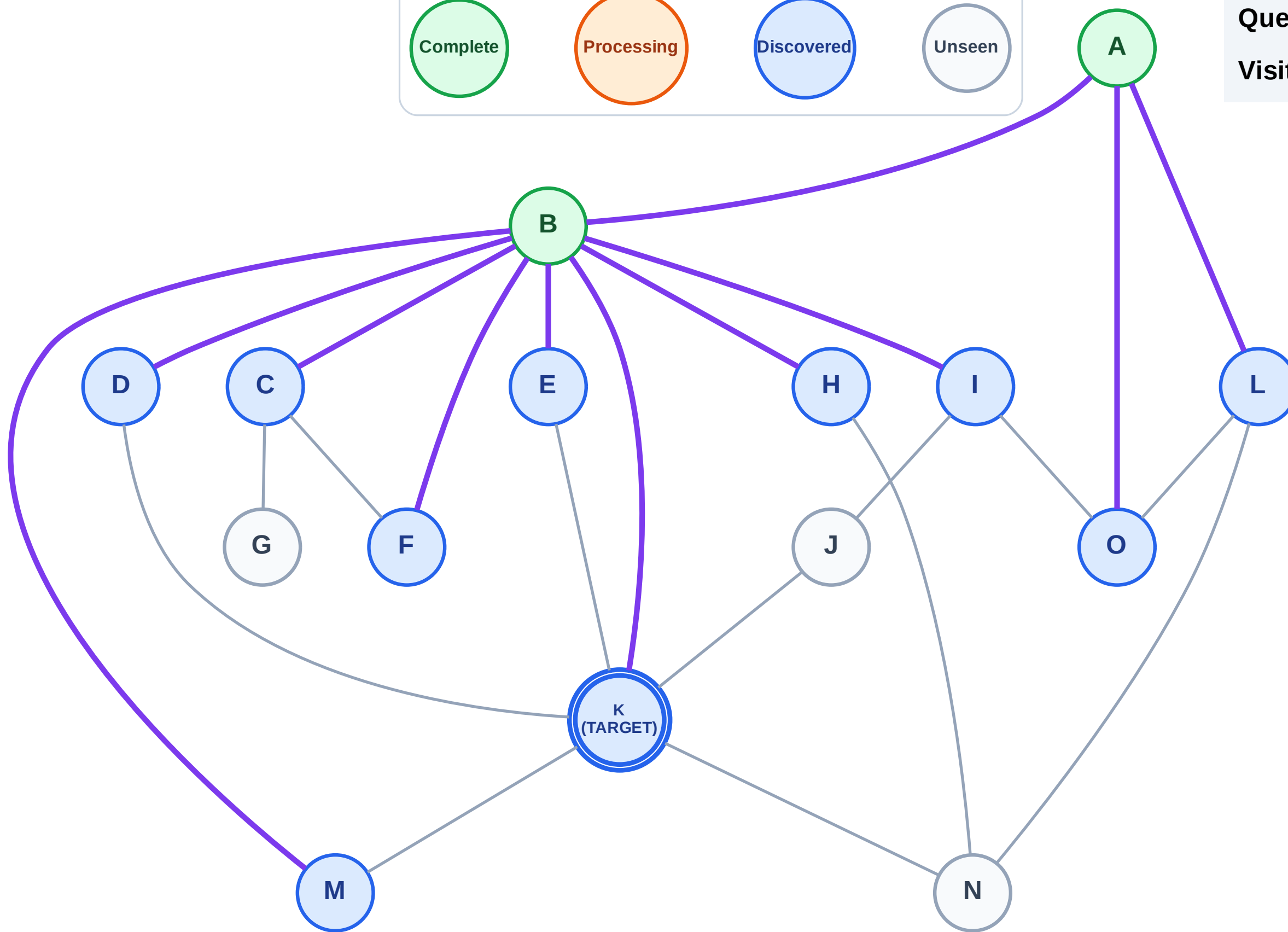
Step 5: Discover C, D, E, F, H, I, K, M from B

Legend



Queue: L → O → C → D → E → F → H → I → K → M

Visited: A, B



BFS Traversal

Step 6: Process node L

Legend

Complete

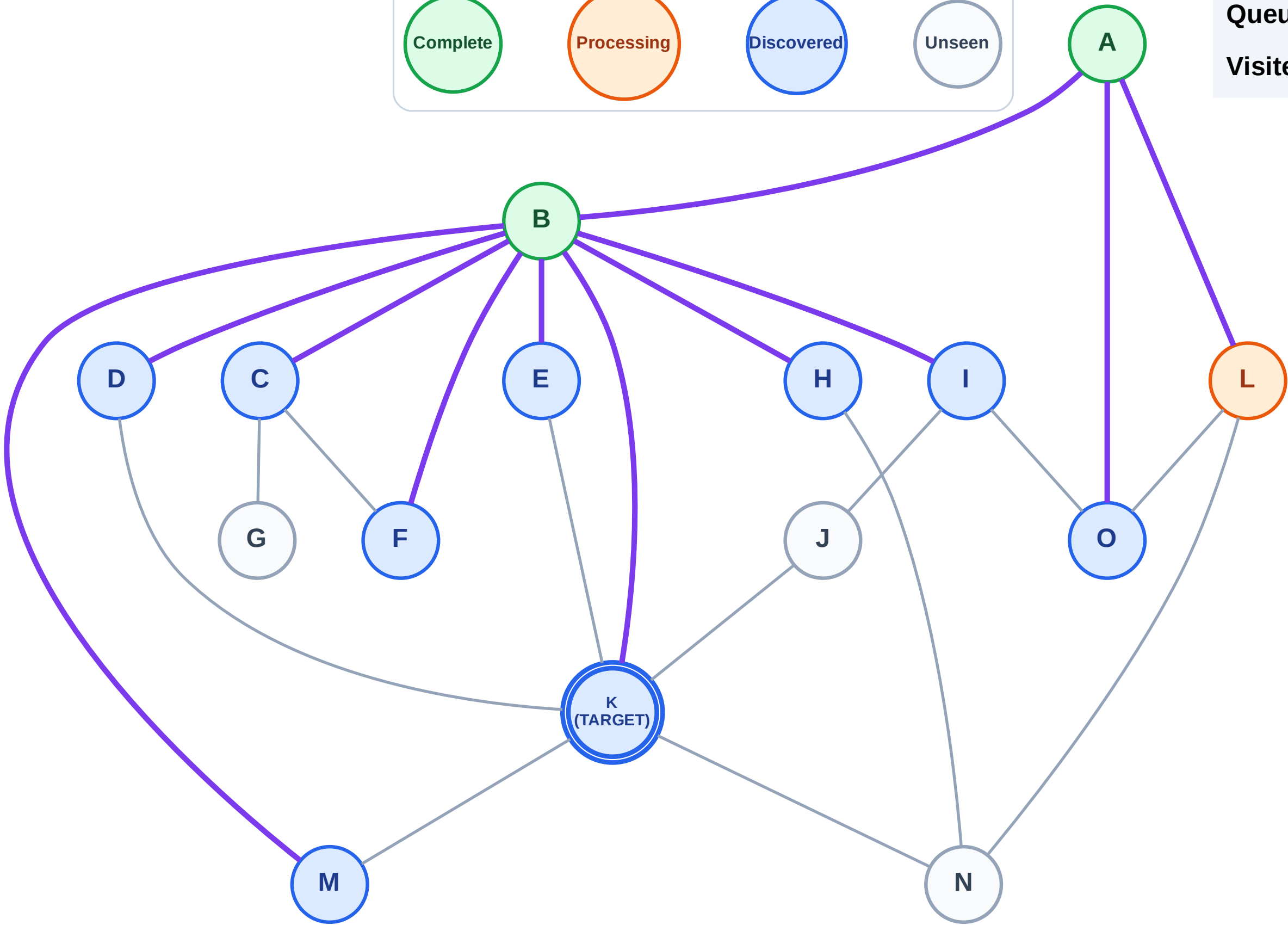
Processing

Discovered

Unseen

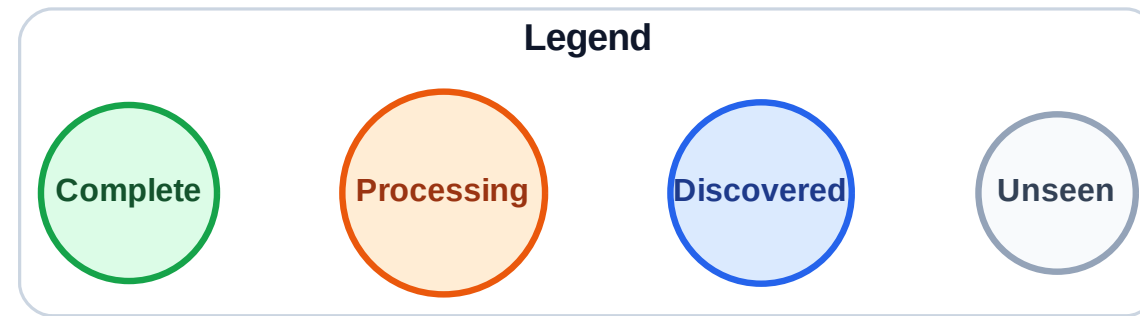
Queue: O → C → D → E → F → H → I → K → M

Visited: A, B



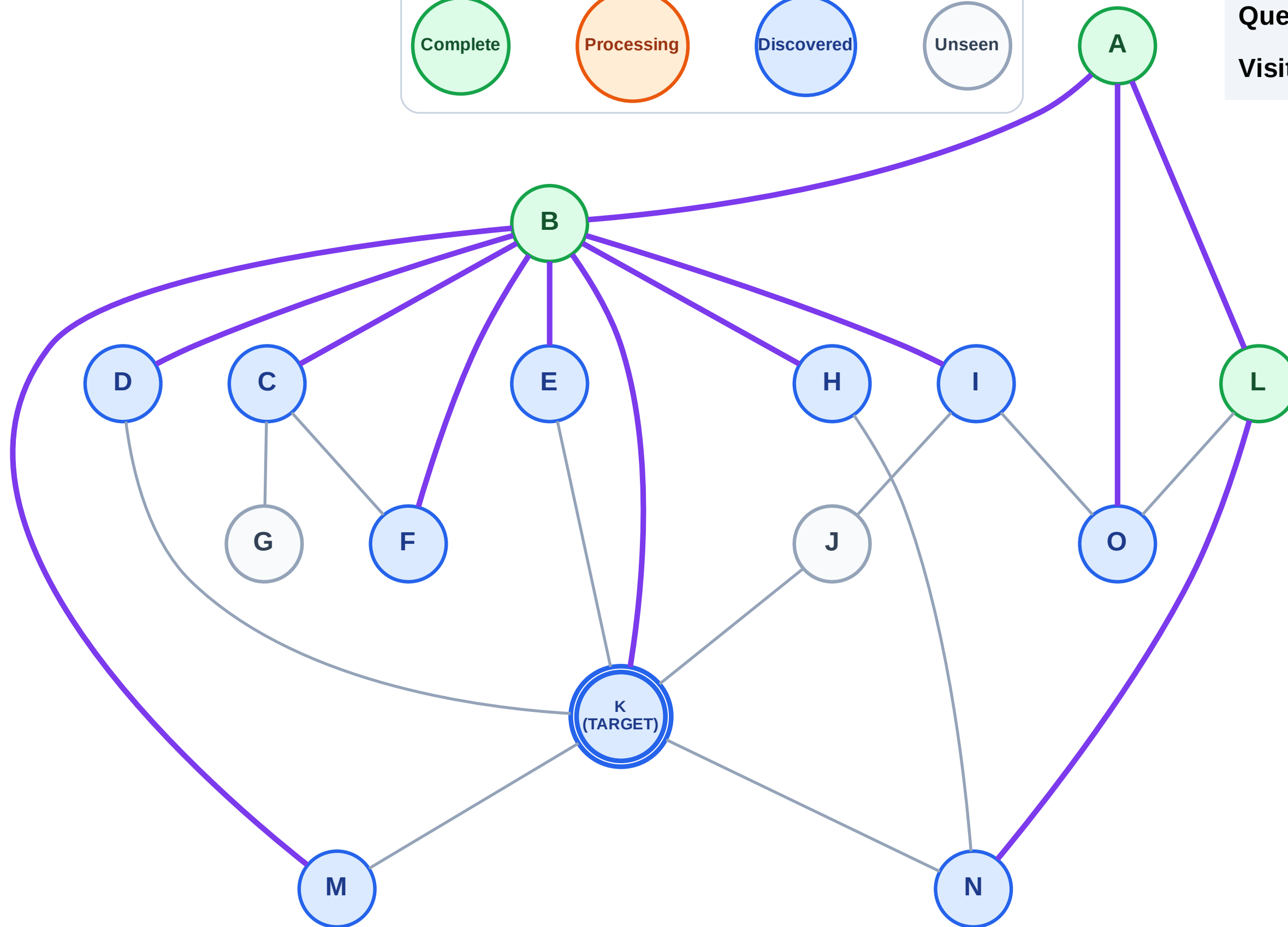
BFS Traversal

Step 7: Discover N from L



Queue: O → C → D → E → F → H → I → K → M → N

Visited: A, B, L



BFS Traversal

Step 8: Process node O

Legend

Complete

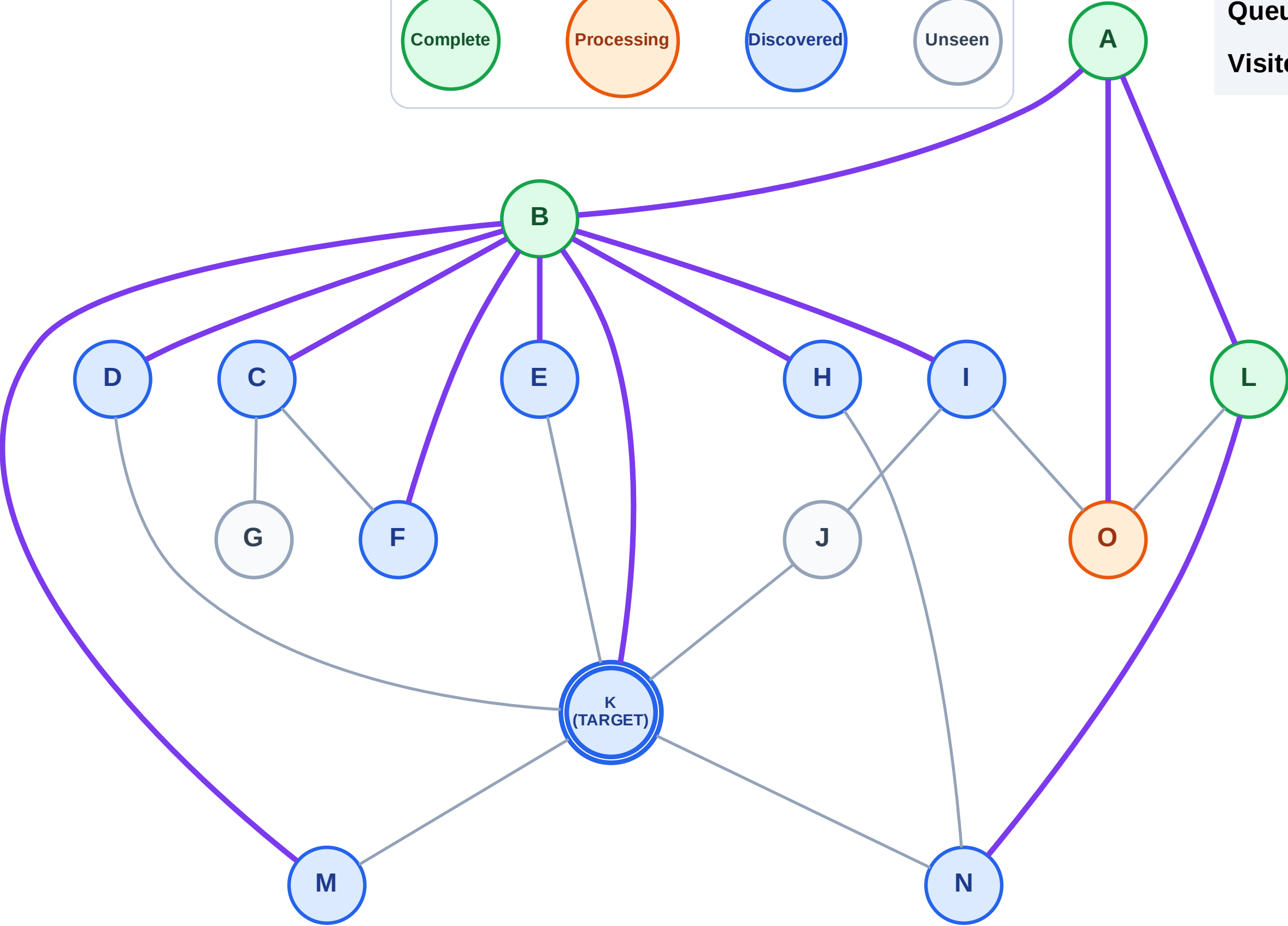
Processing

Discovered

Unseen

Queue: C → D → E → F → H → I → K → M → N

Visited: A, B, L



BFS Traversal

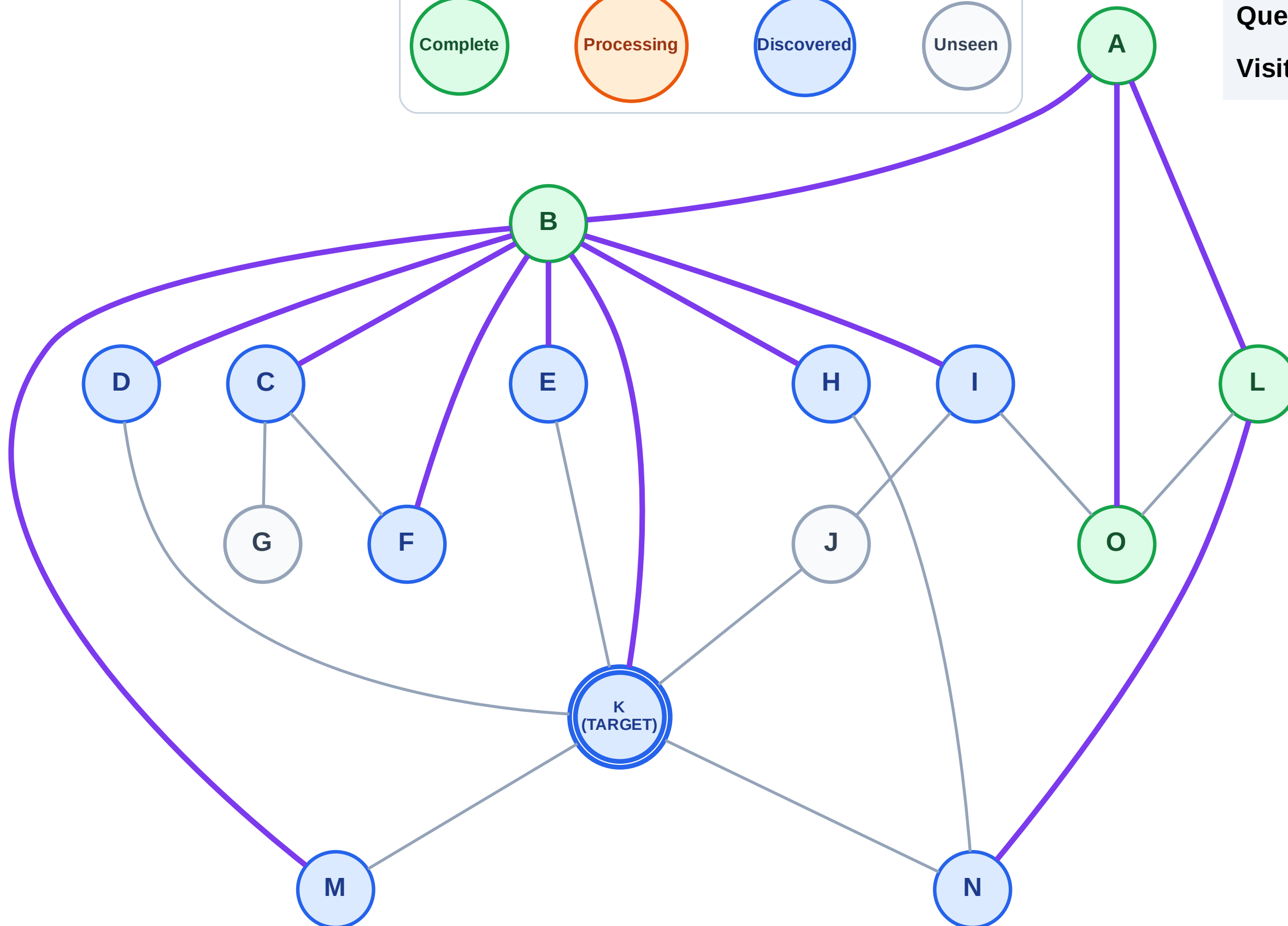
Step 9: No new neighbors from O

Legend



Queue: C → D → E → F → H → I → K → M → N

Visited: A, B, L, O



BFS Traversal

Step 10: Process node C

Legend

Complete

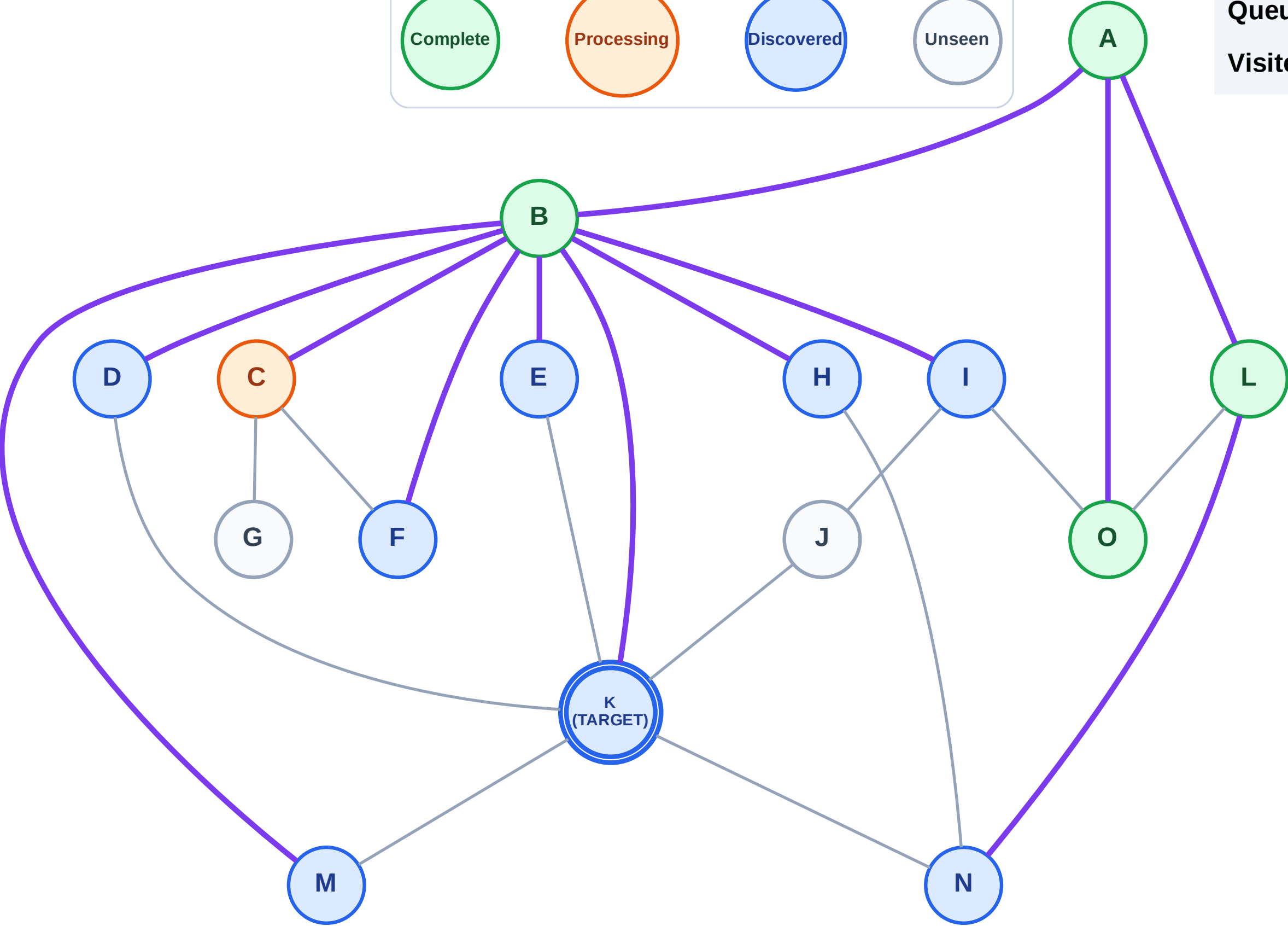
Processing

Discovered

Unseen

Queue: D → E → F → H → I → K → M → N

Visited: A, B, L, O



BFS Traversal

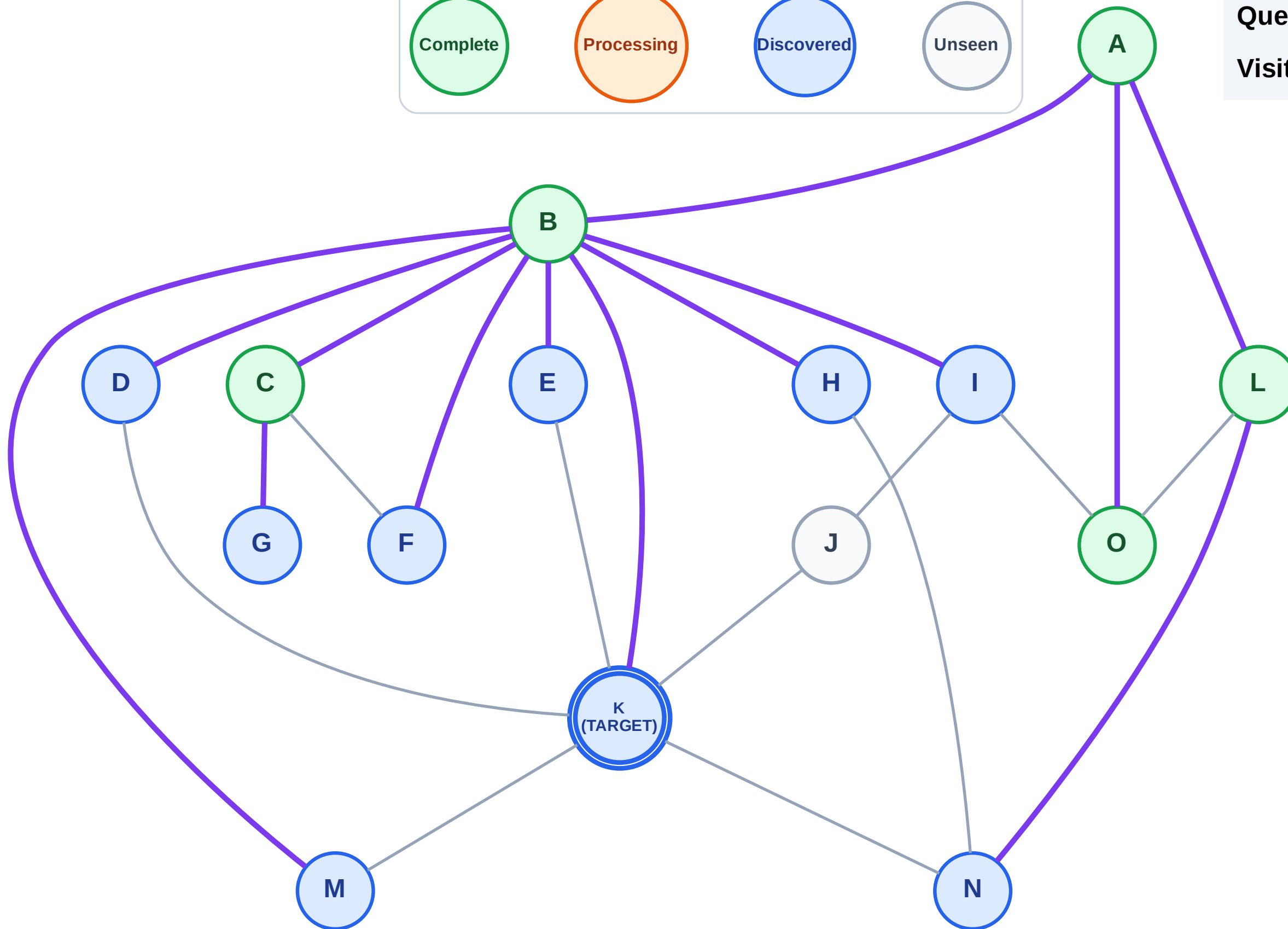
Step 11: Discover G from C

Legend



Queue: D → E → F → H → I → K → M → N → G

Visited: A, B, C, L, O



BFS Traversal

Step 12: Process node D

Legend

Complete

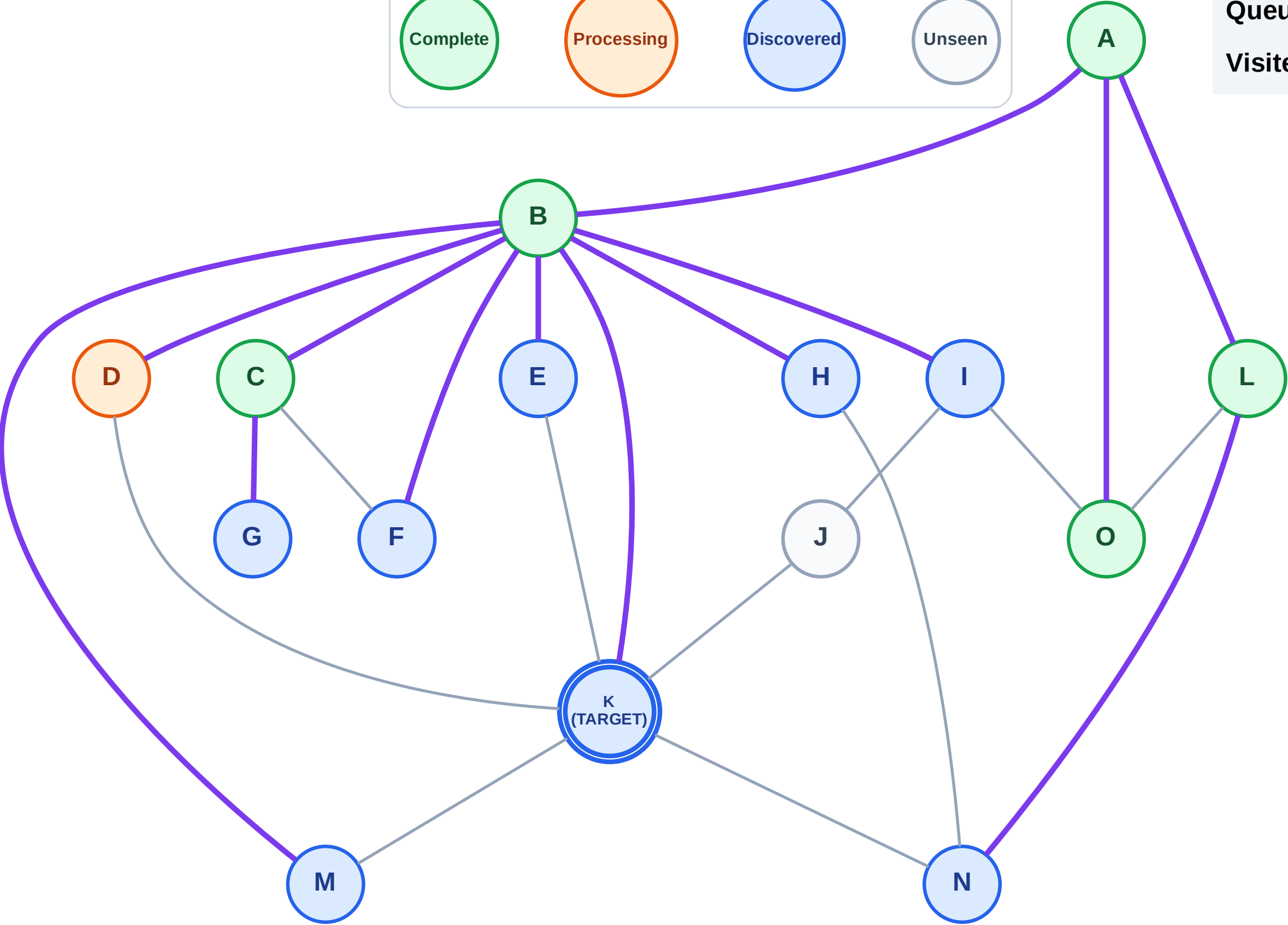
Processing

Discovered

Unseen

Queue: E → F → H → I → K → M → N → G

Visited: A, B, C, L, O



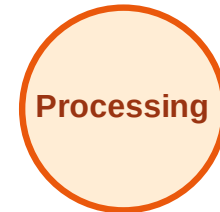
BFS Traversal

Step 13: No new neighbors from D

Legend



Complete



Processing



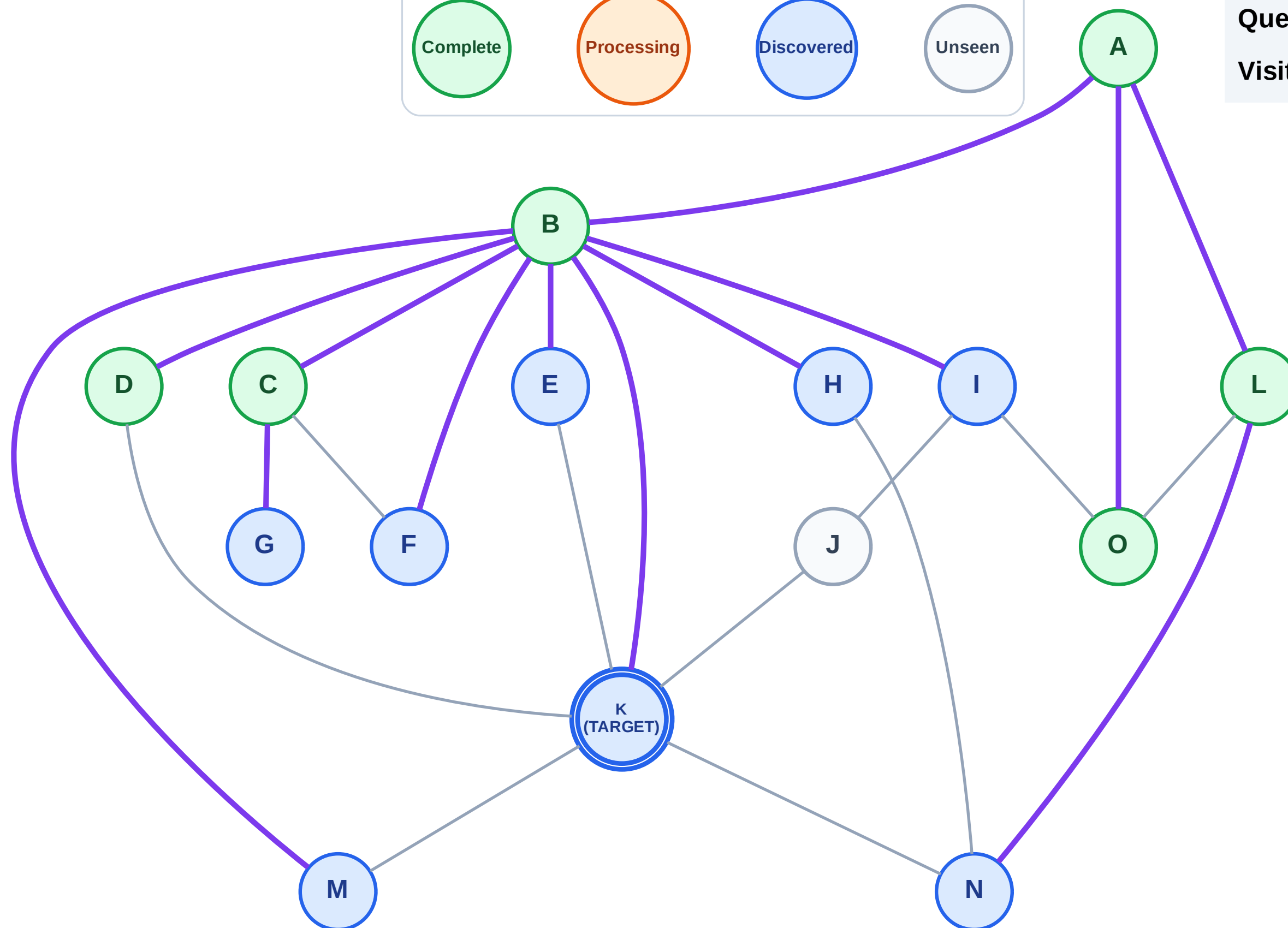
Discovered



Unseen

Queue: E → F → H → I → K → M → N → G

Visited: A, B, C, D, L, O



BFS Traversal

Step 14: Process node E

Legend

Complete

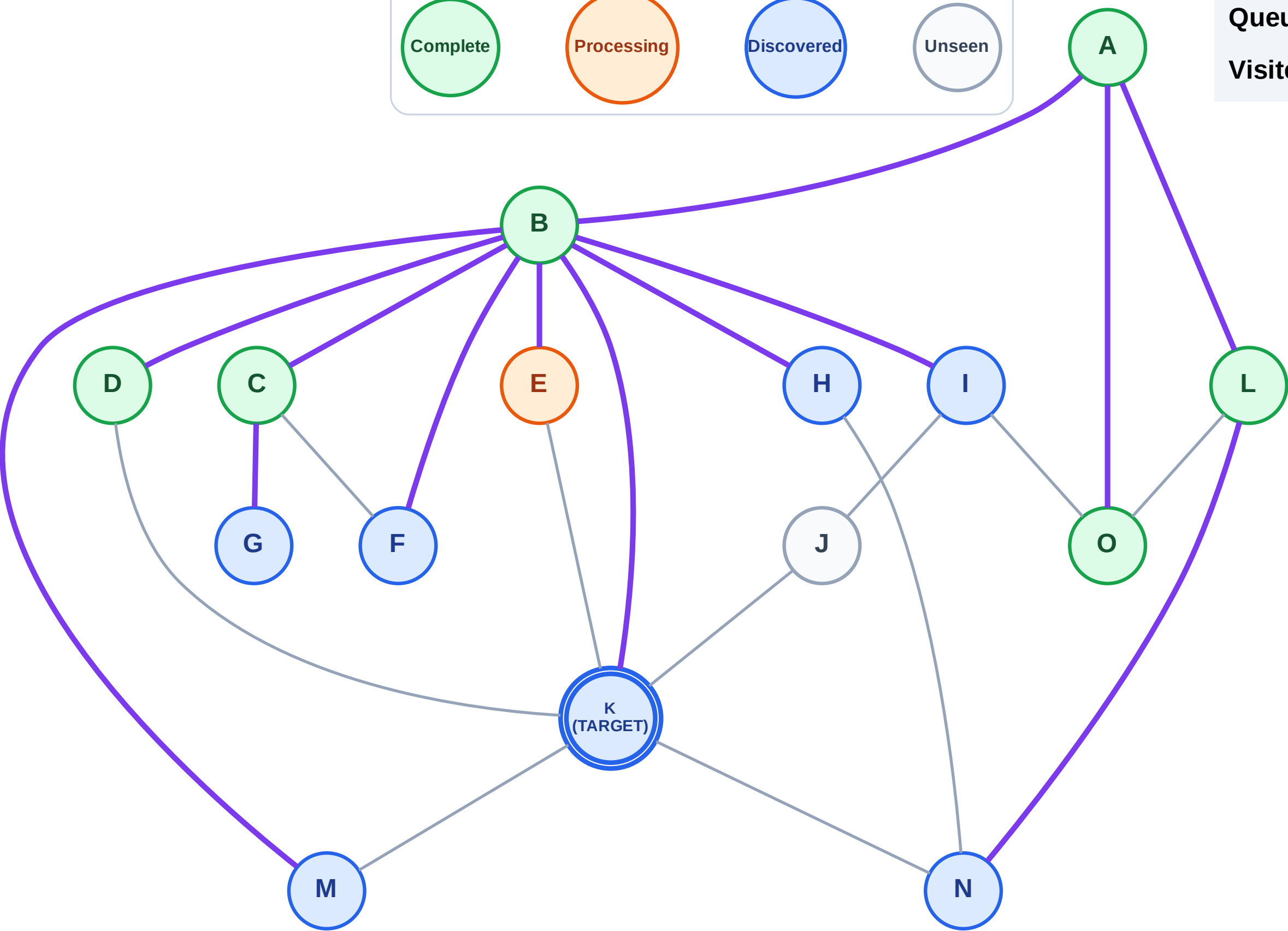
Processing

Discovered

Unseen

Queue: F → H → I → K → M → N → G

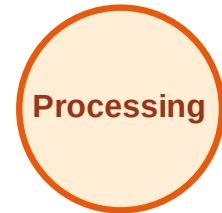
Visited: A, B, C, D, L, O



BFS Traversal

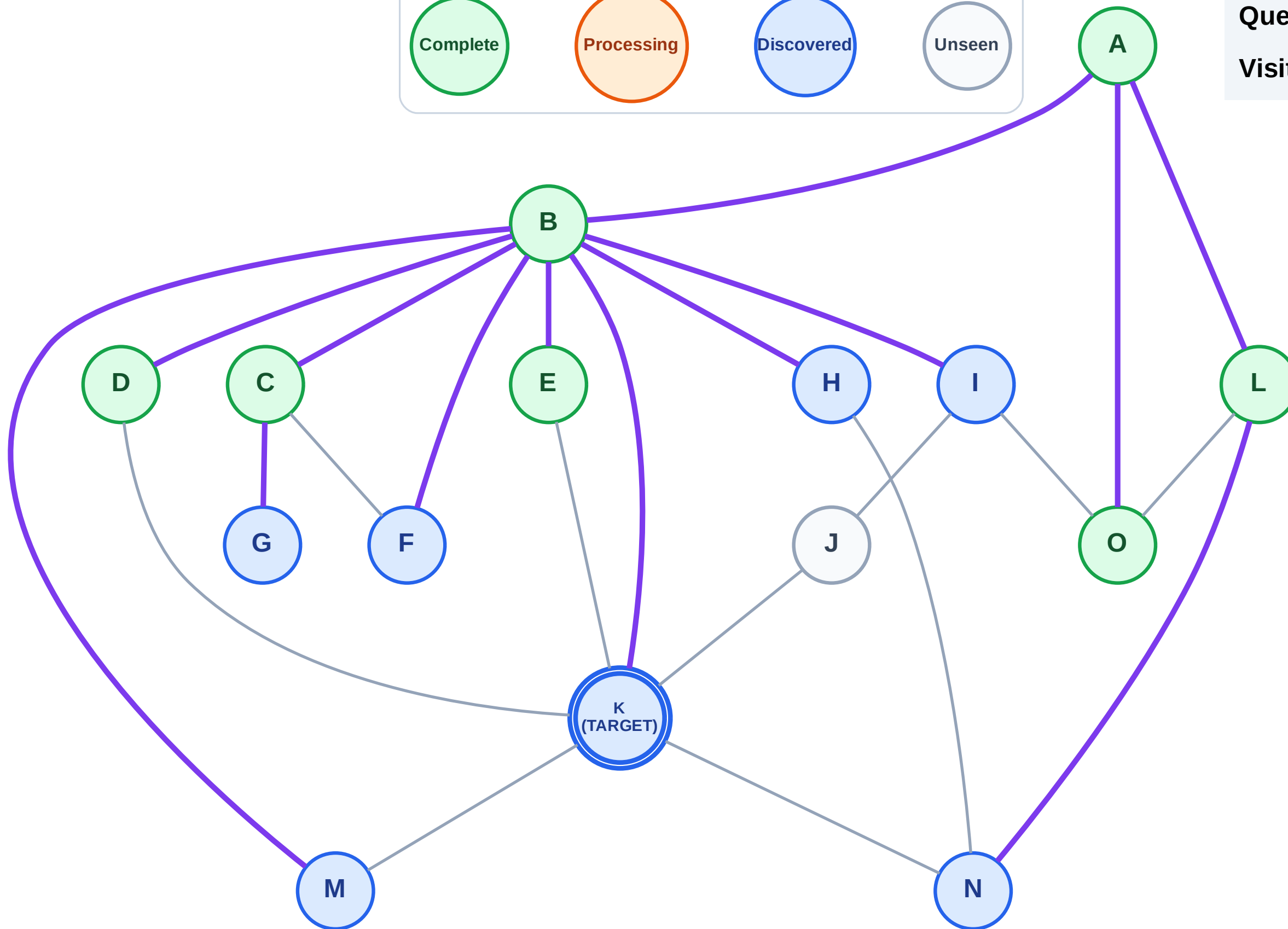
Step 15: No new neighbors from E

Legend



Queue: F → H → I → K → M → N → G

Visited: A, B, C, D, E, L, O



BFS Traversal

Step 16: Process node F

Legend

Complete

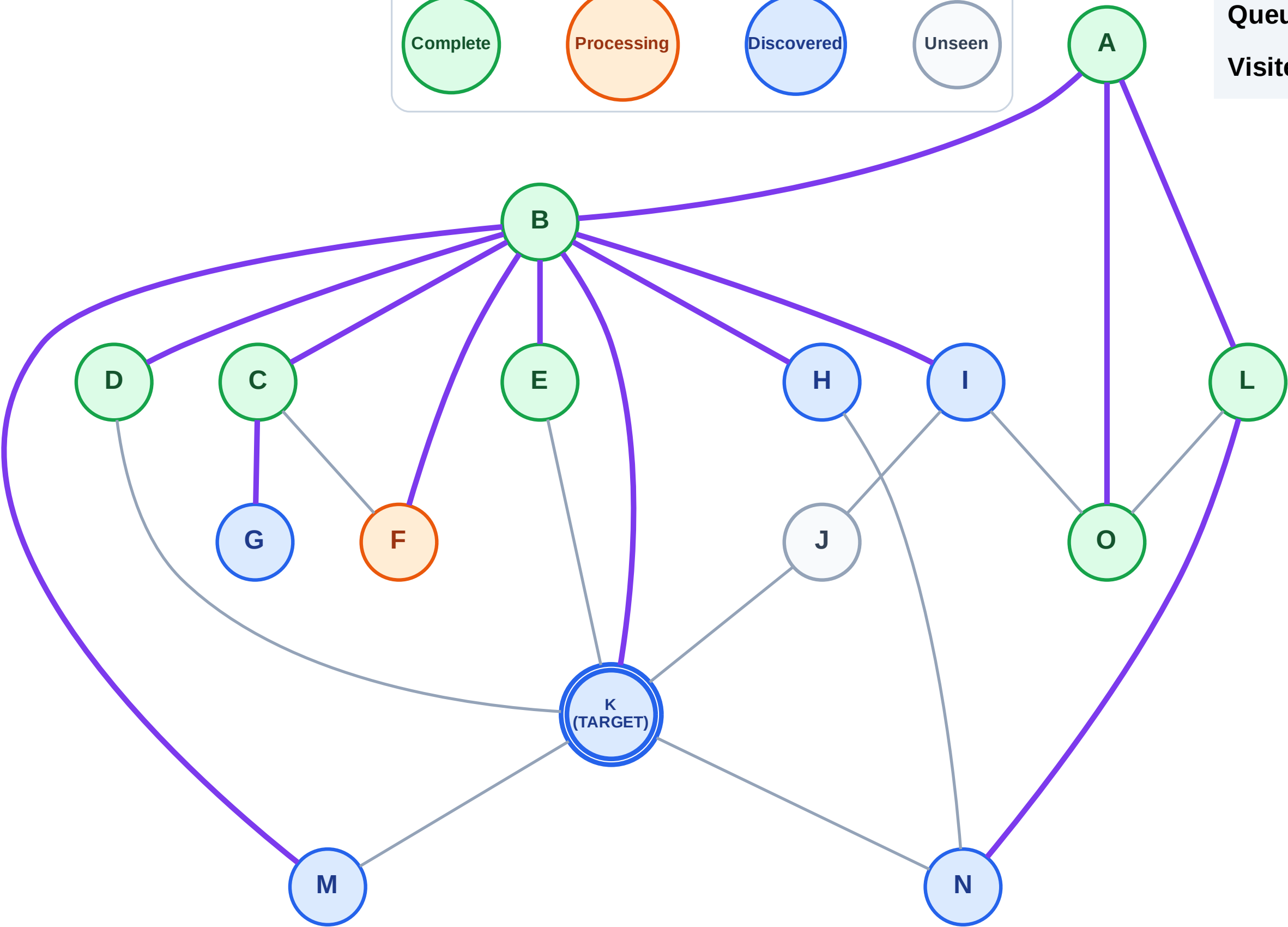
Processing

Discovered

Unseen

Queue: H → I → K → M → N → G

Visited: A, B, C, D, E, L, O



BFS Traversal

Step 17: No new neighbors from F

Legend

Complete

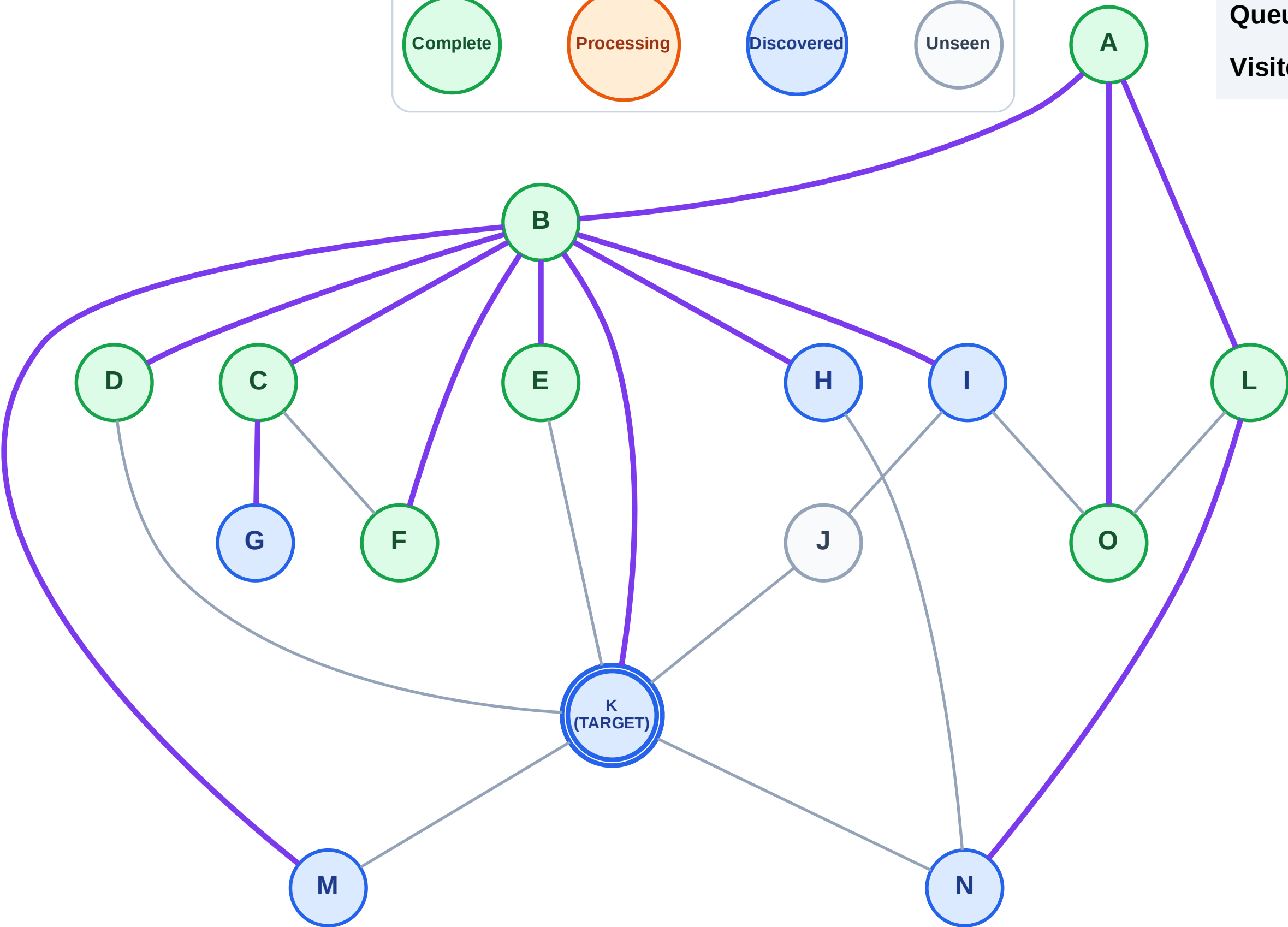
Processing

Discovered

Unseen

Queue: H → I → K → M → N → G

Visited: A, B, C, D, E, F, L, O



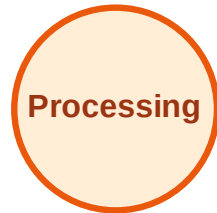
BFS Traversal

Step 18: Process node H

Legend



Complete



Processing



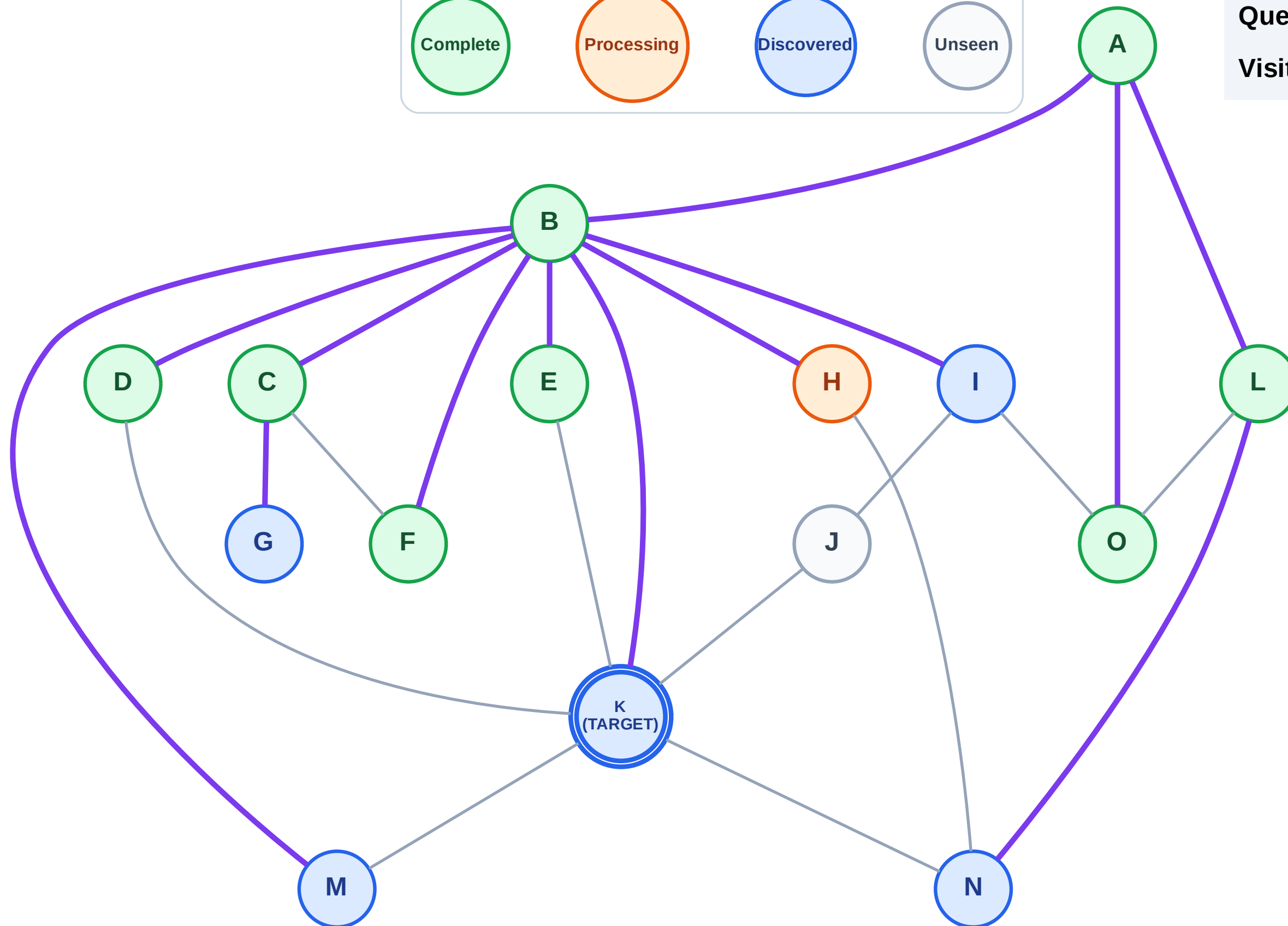
Discovered



Unseen

Queue: I → K → M → N → G

Visited: A, B, C, D, E, F, L, O



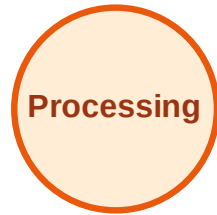
BFS Traversal

Step 19: No new neighbors from H

Legend



Complete



Processing



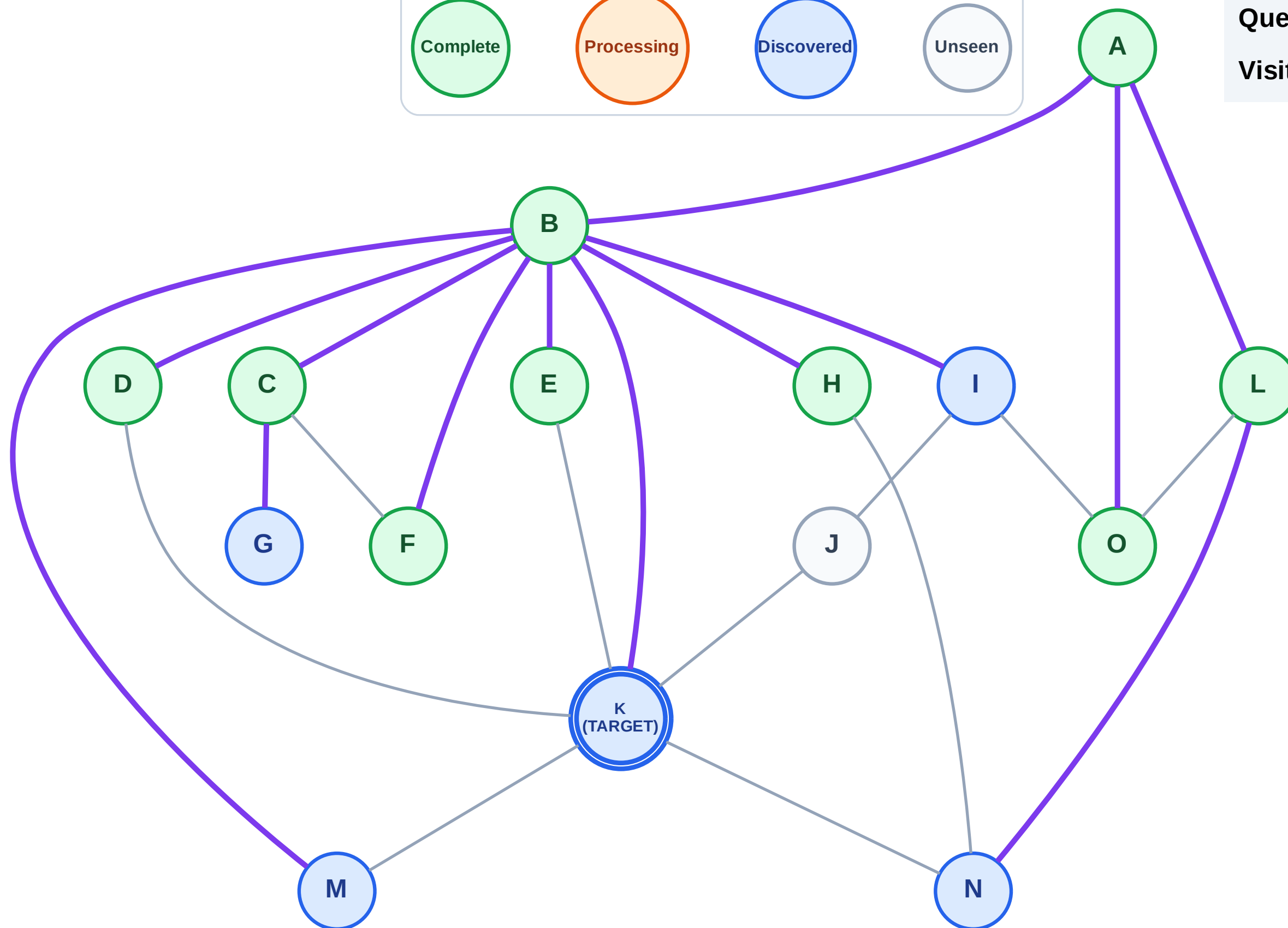
Discovered



Unseen

Queue: I → K → M → N → G

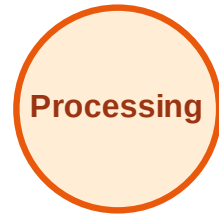
Visited: A, B, C, D, E, F, H, L, O



BFS Traversal

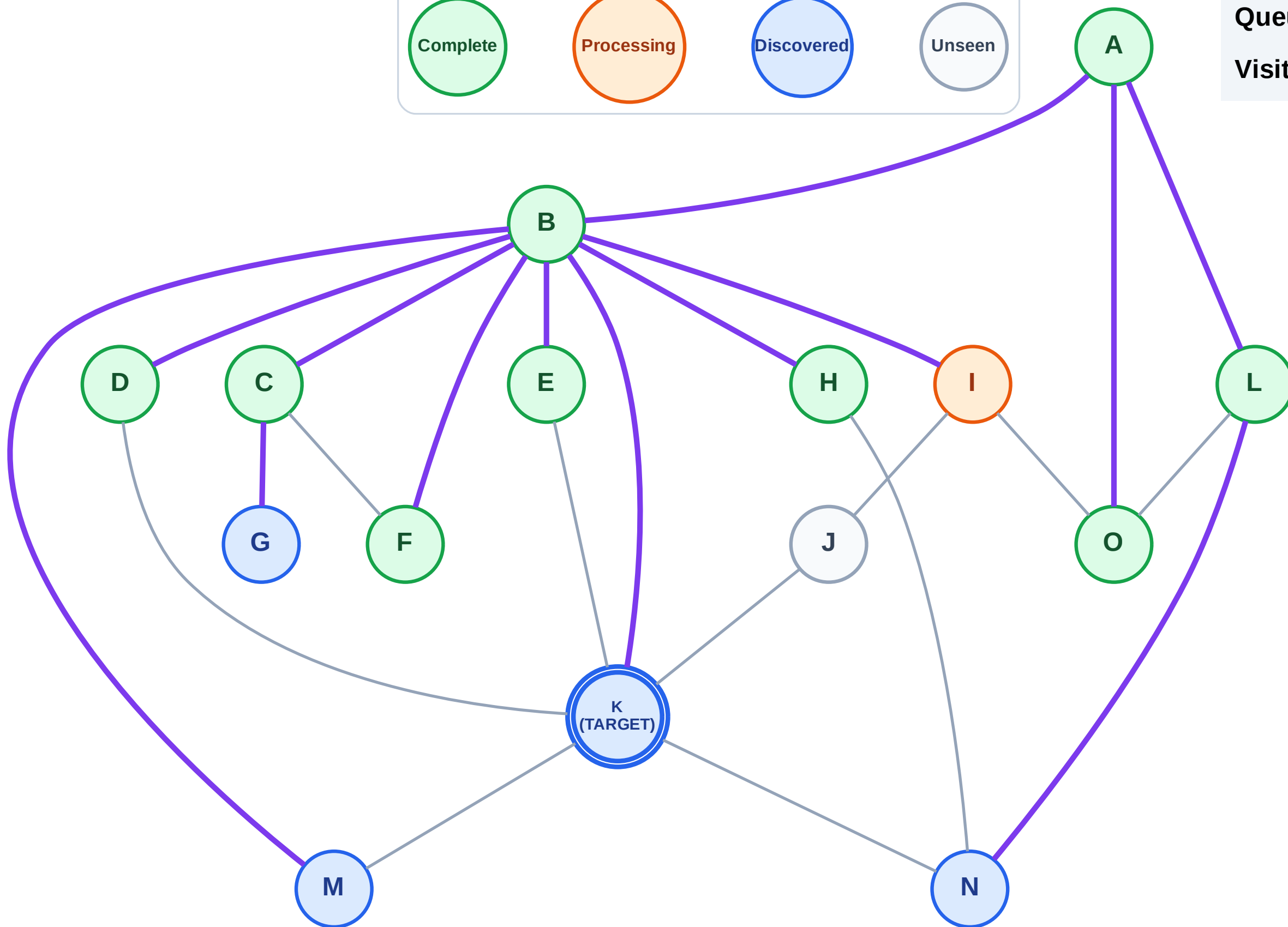
Step 20: Process node I

Legend



Queue: K → M → N → G

Visited: A, B, C, D, E, F, H, L, O



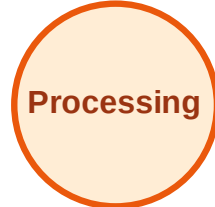
BFS Traversal

Step 21: Discover J from I

Legend



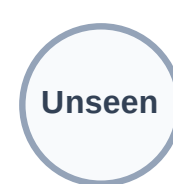
Complete



Processing



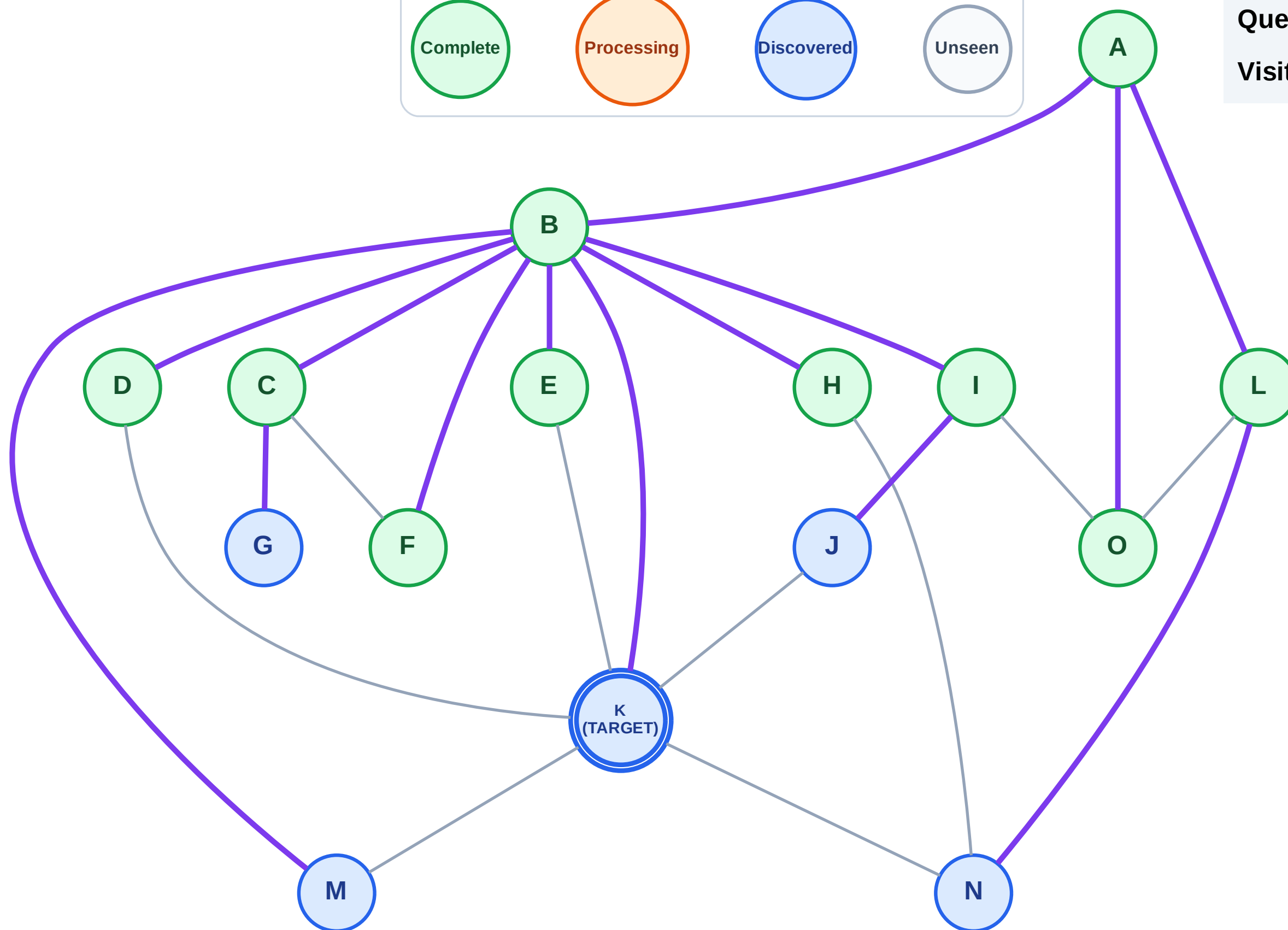
Discovered



Unseen

Queue: K → M → N → G → J

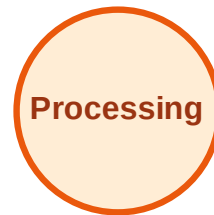
Visited: A, B, C, D, E, F, H, I, L, O



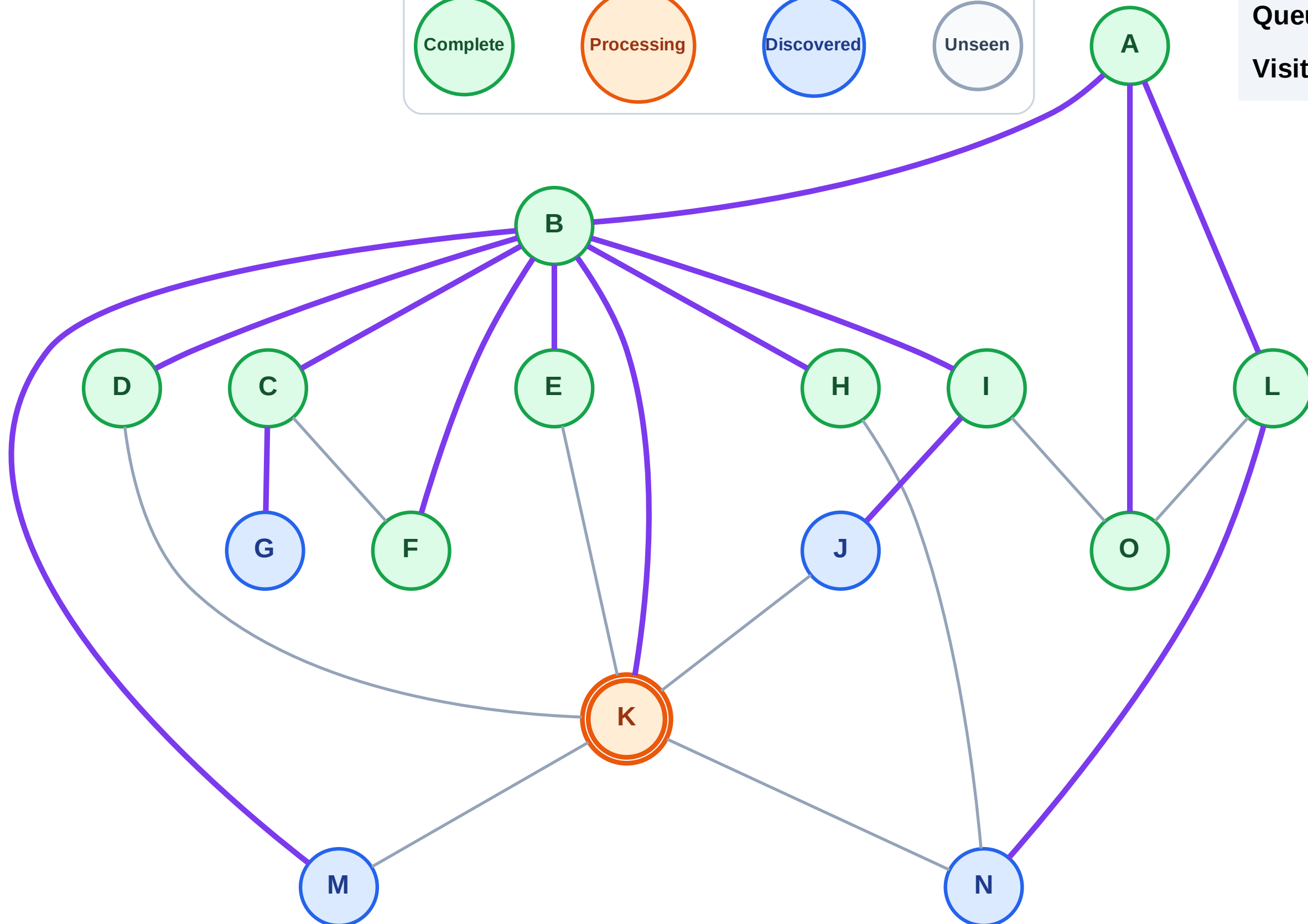
BFS Traversal

Step 22: Process node K

Legend



Queue: M → N → G → J
Visited: A, B, C, D, E, F, H, I, L, O



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



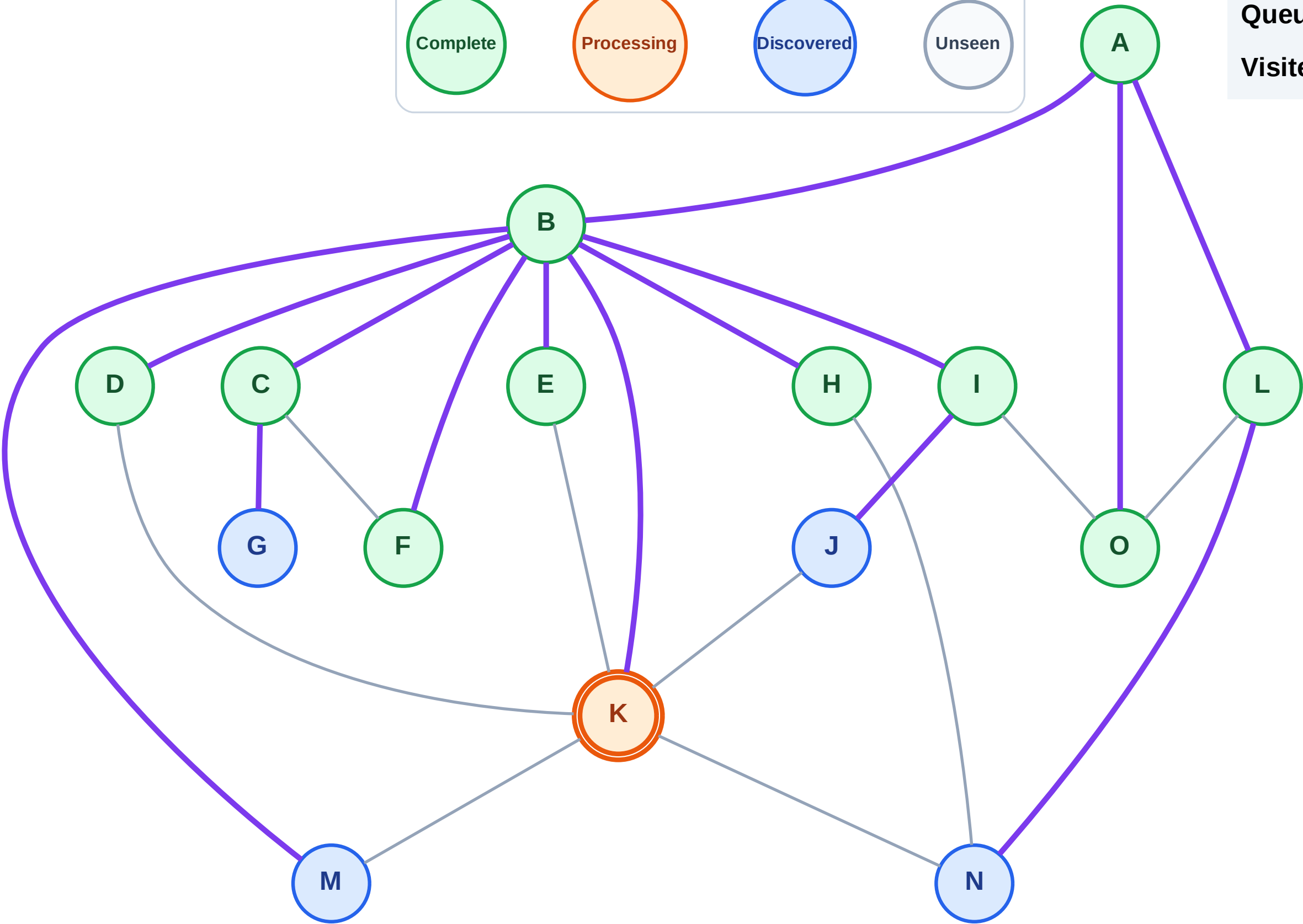
Discovered



Unseen

Queue: M → N → G → J

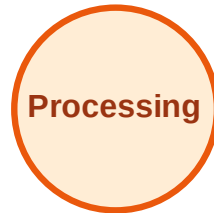
Visited: A, B, C, D, E, F, H, I, K, L, O



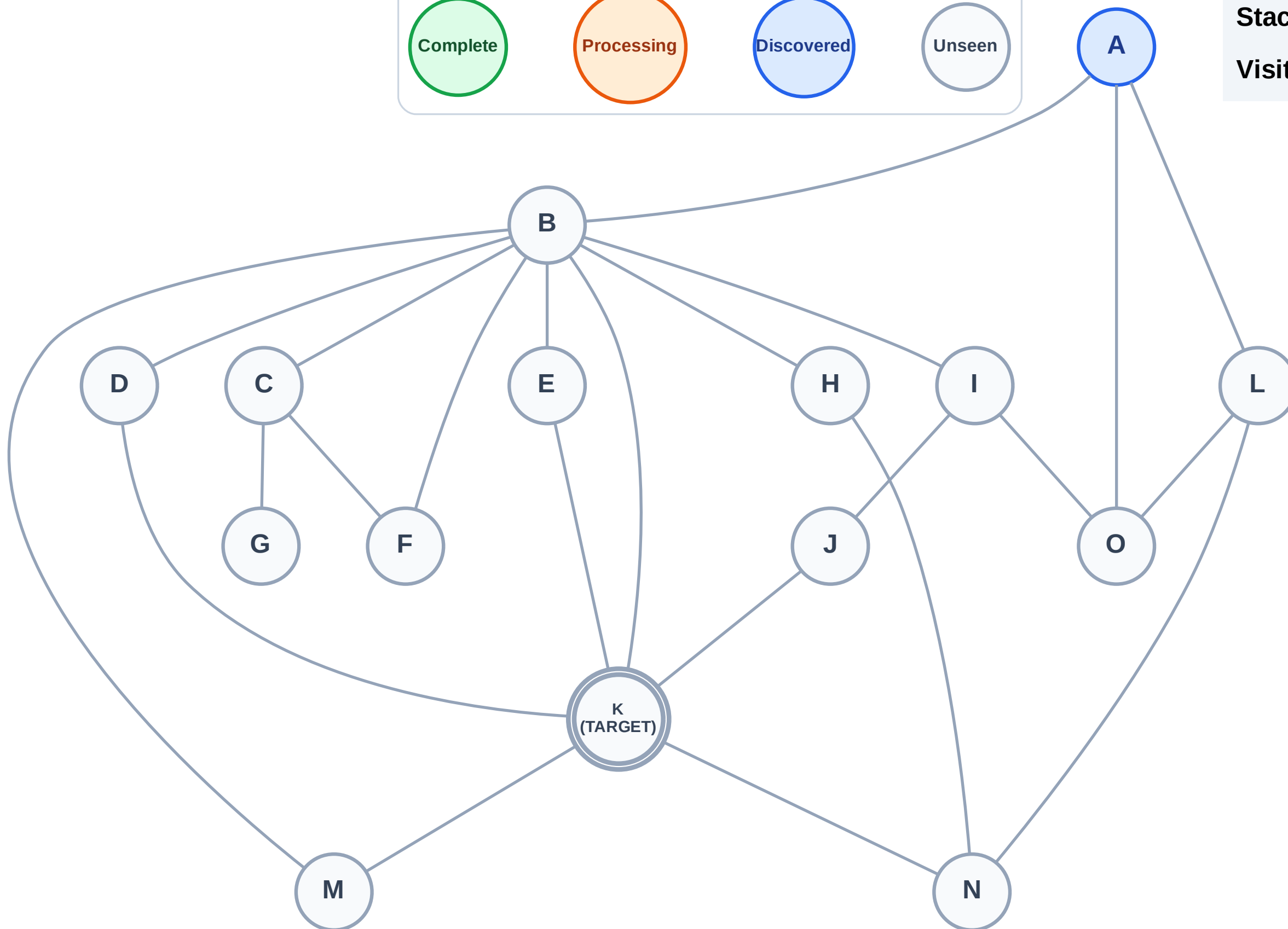
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



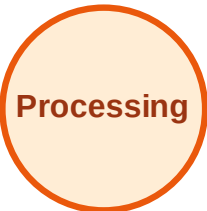
Stack (top at right): A
Visited: (none)



DFS Traversal

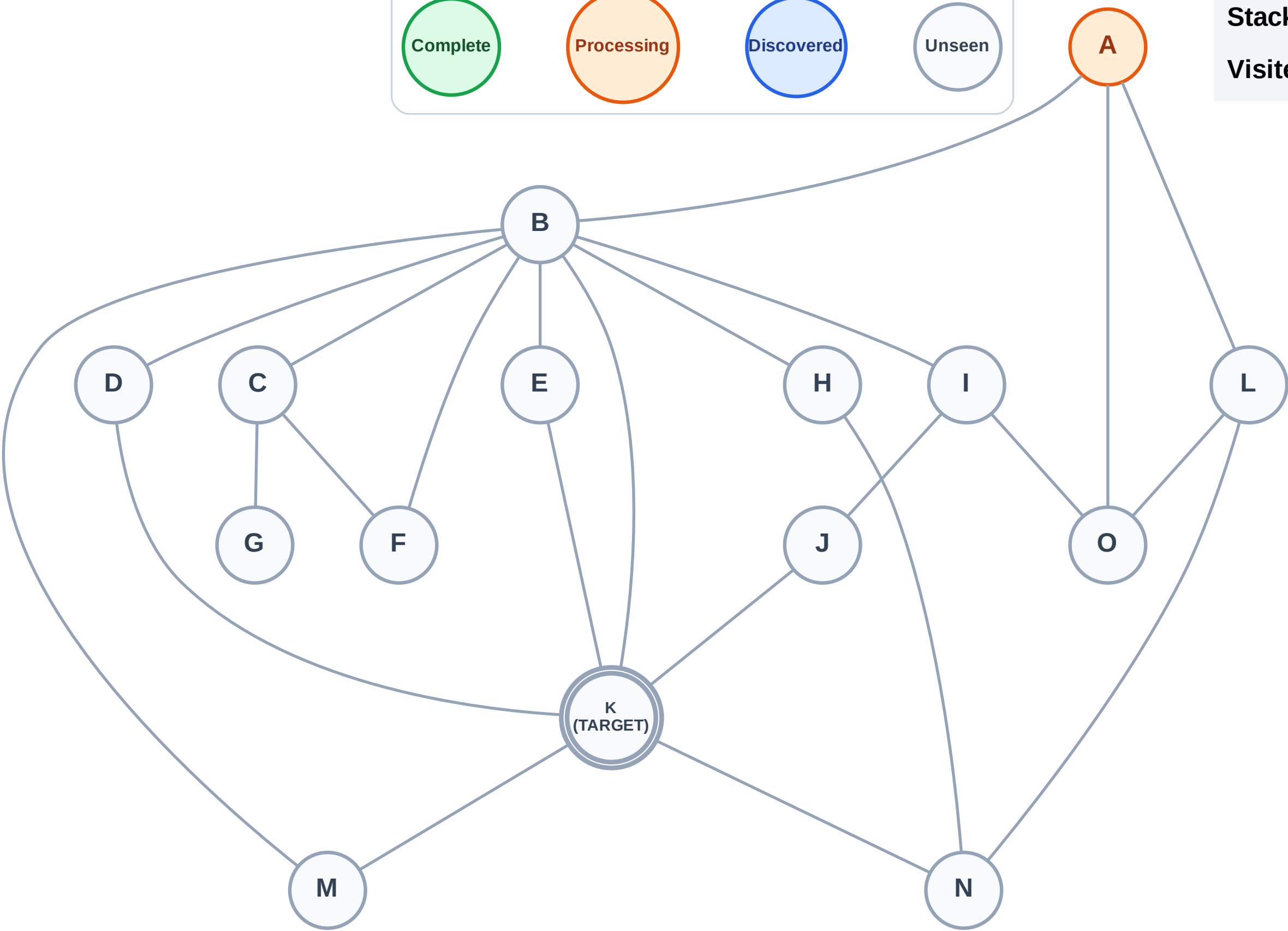
Step 2: Process node A

Legend



Stack (top at right): (empty)

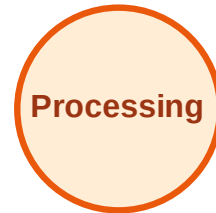
Visited: (none)



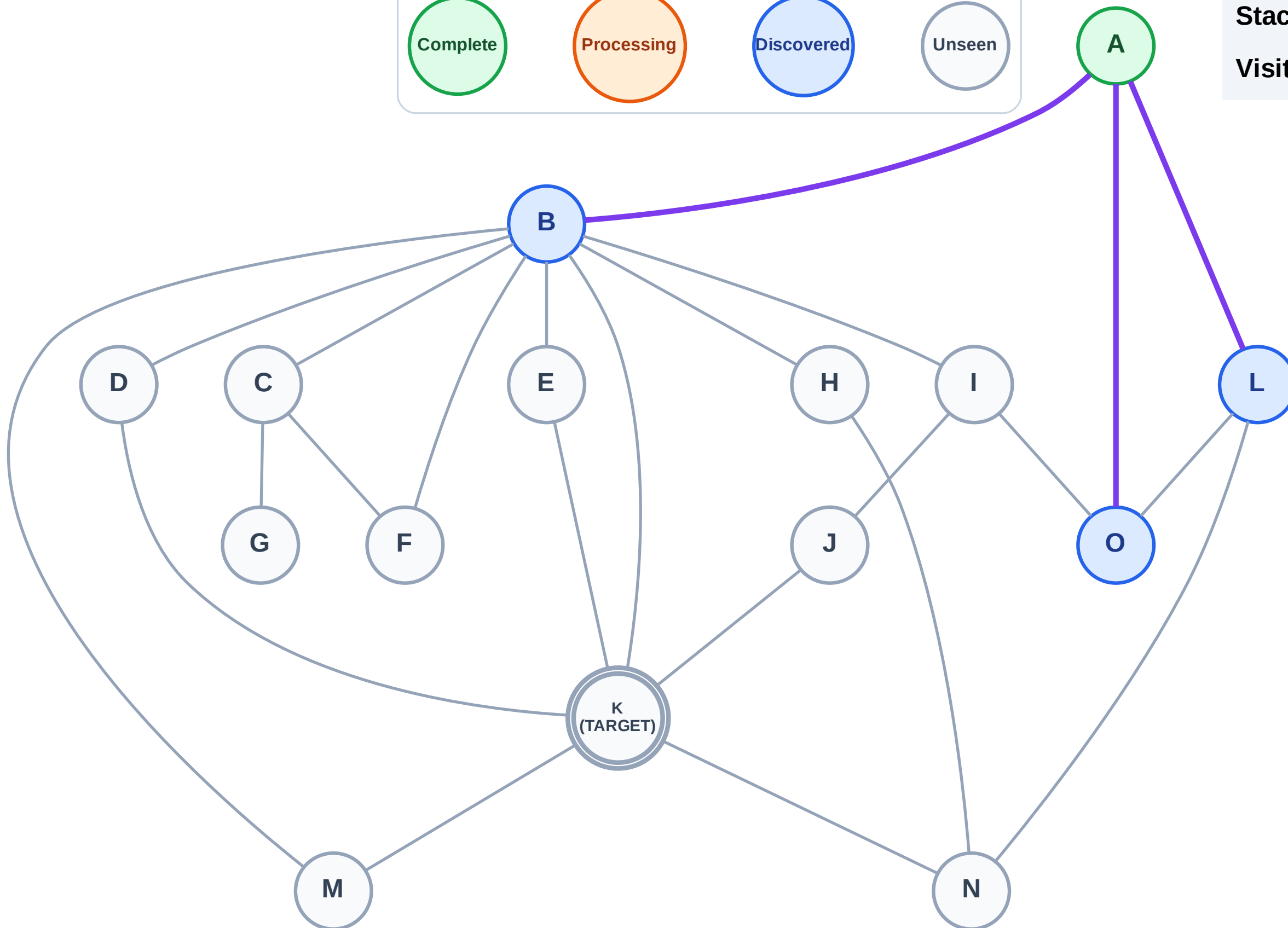
DFS Traversal

Step 3: Discover O, L, B from A

Legend



Stack (top at right): O | L | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

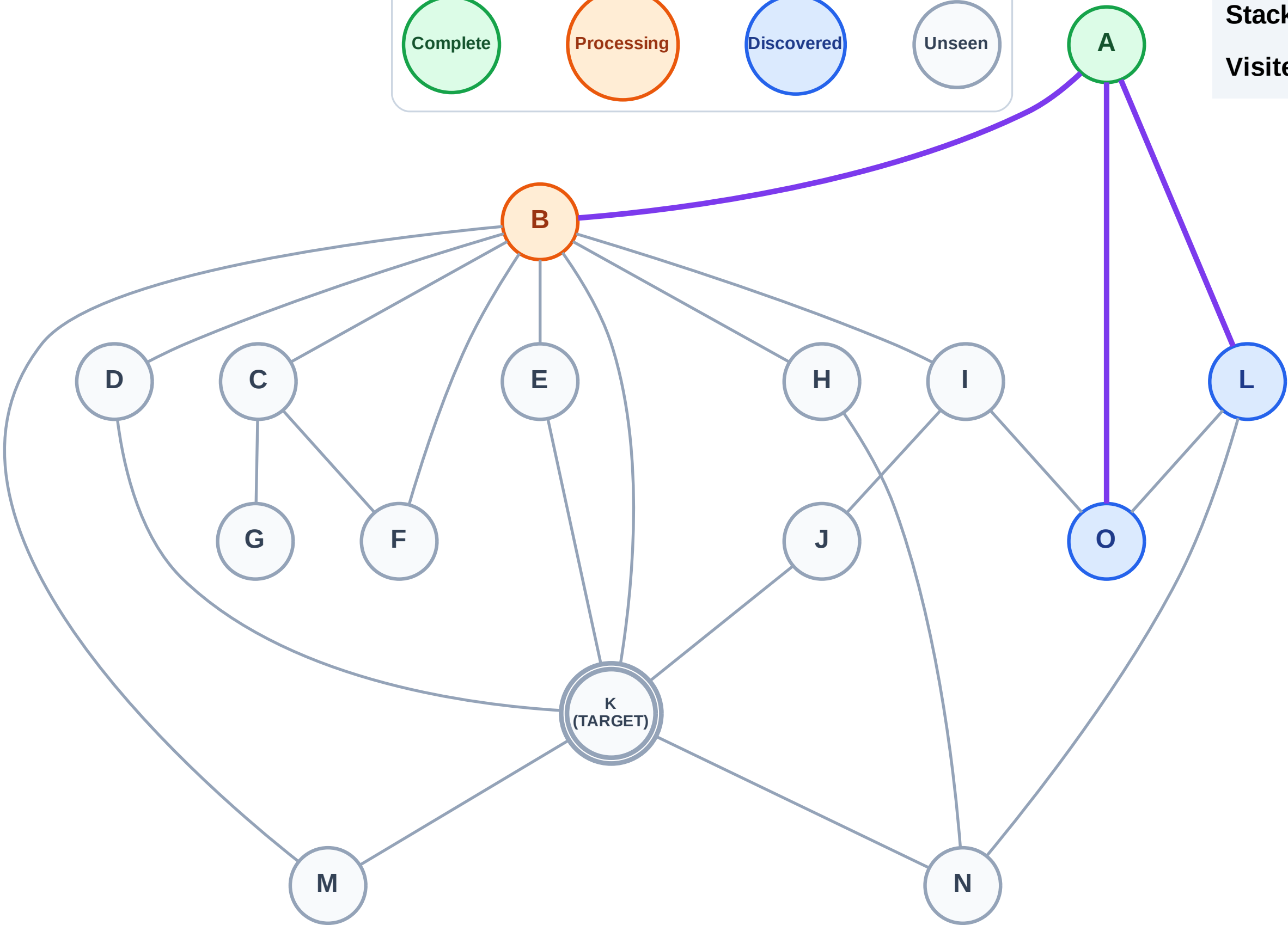
Complete

Processing

Discovered

Unseen

Stack (top at right): O | L
Visited: A



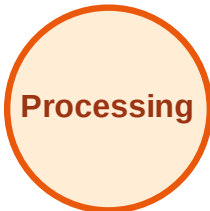
DFS Traversal

Step 5: Discover M, K, I, H, F, E, D, C from B

Legend



Complete



Processing



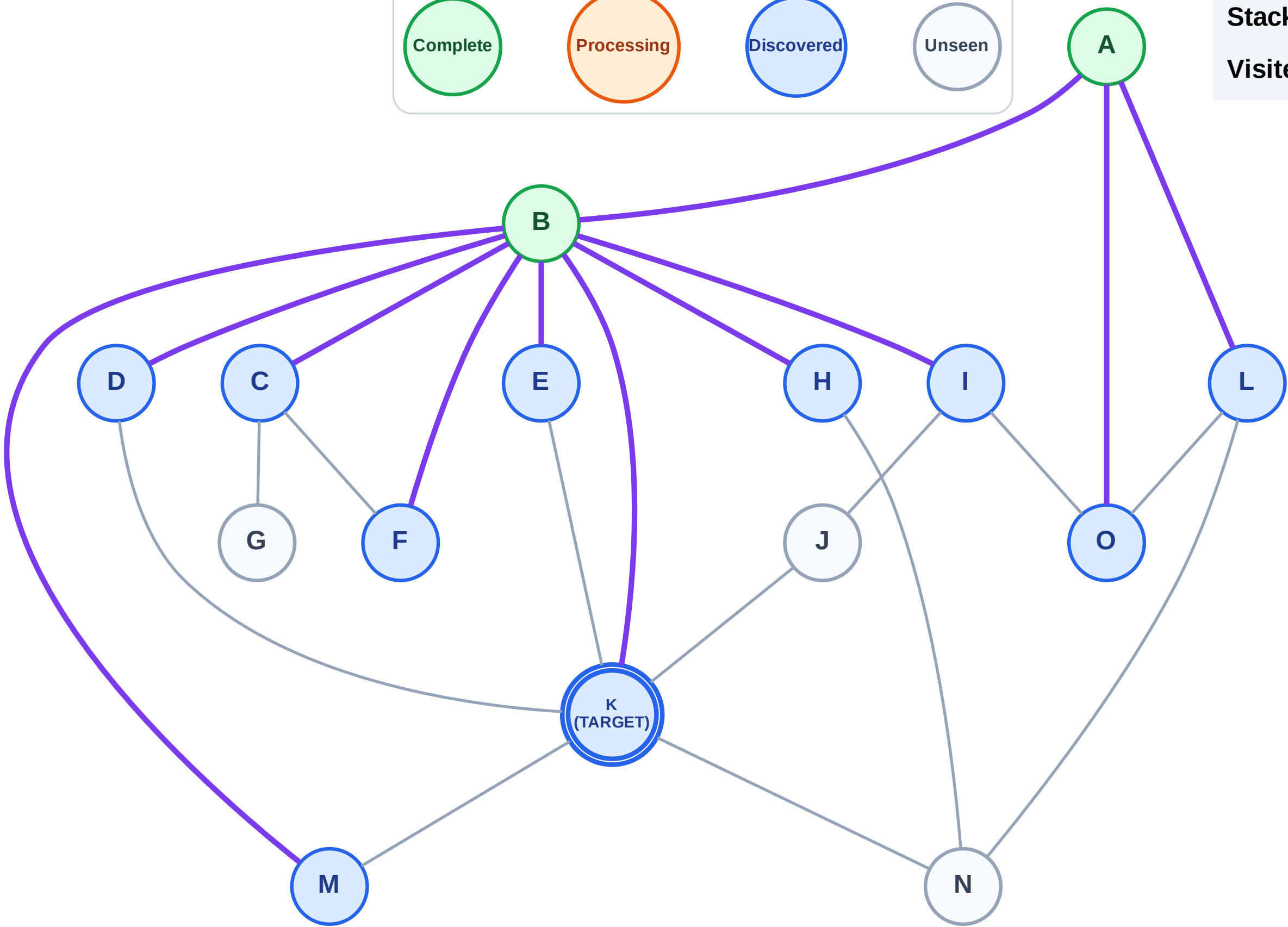
Discovered



Unseen

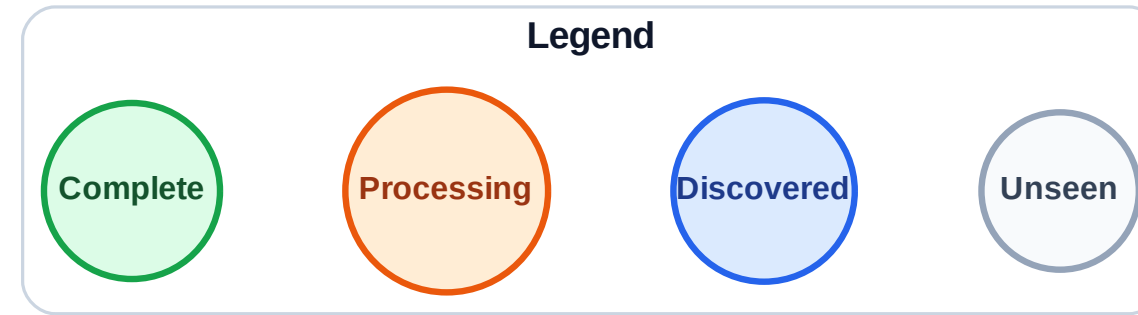
Stack (top at right): O | L | M | K | I | H | F | E | D | C

Visited: A, B



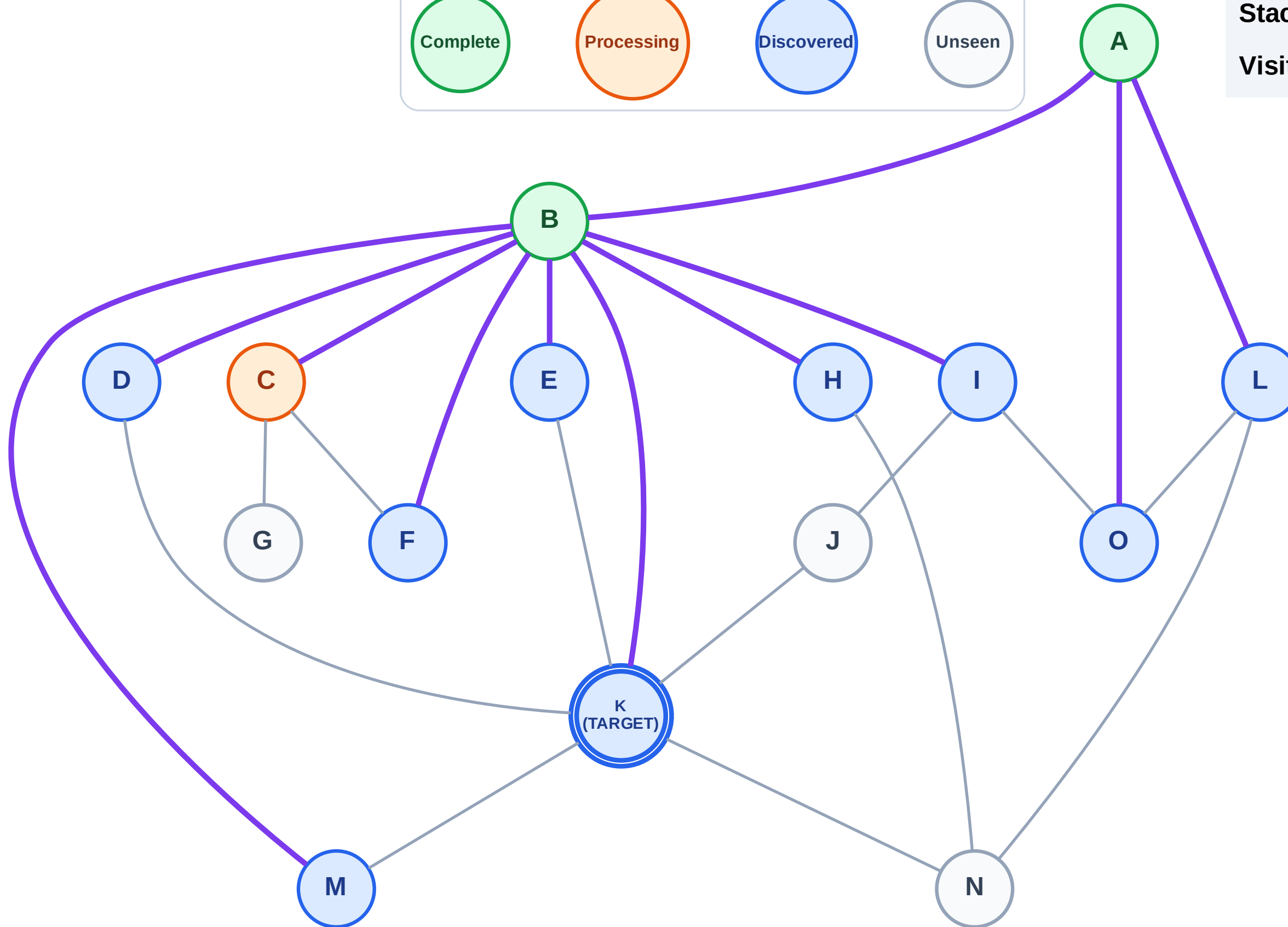
Step 6: Process node C

Page 14 of 14



Stack (top at right): O | L | M | K | I | H | F | E | D

Visited: A, B



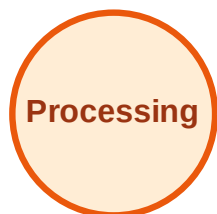
DFS Traversal

Step 7: Discover G from C

Legend



Complete



Processing



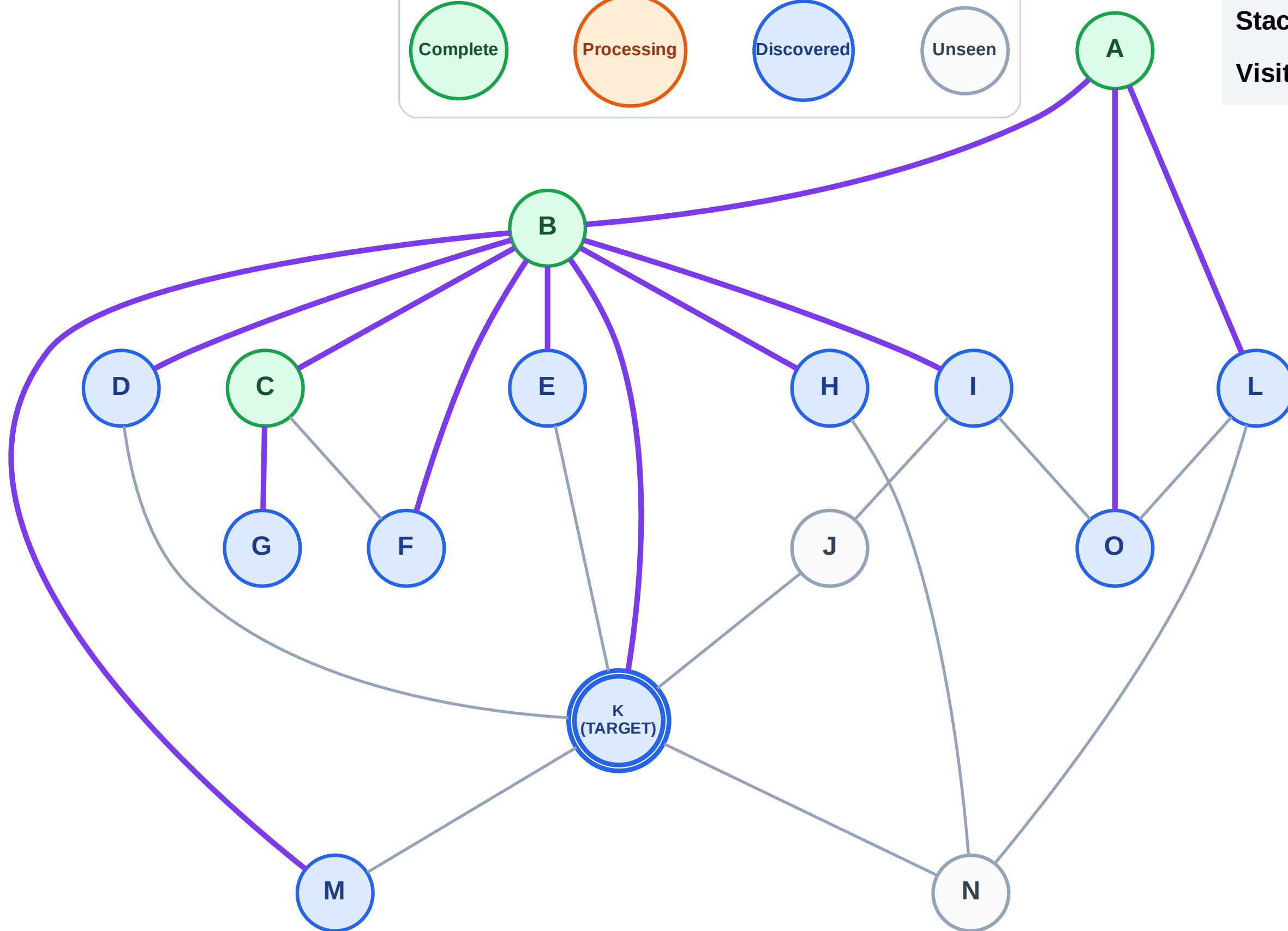
Discovered



Unseen

Stack (top at right): O | L | M | K | I | H | F | E | D | G

Visited: A, B, C



DFS Traversal

Step 8: Process node G

Legend

Complete

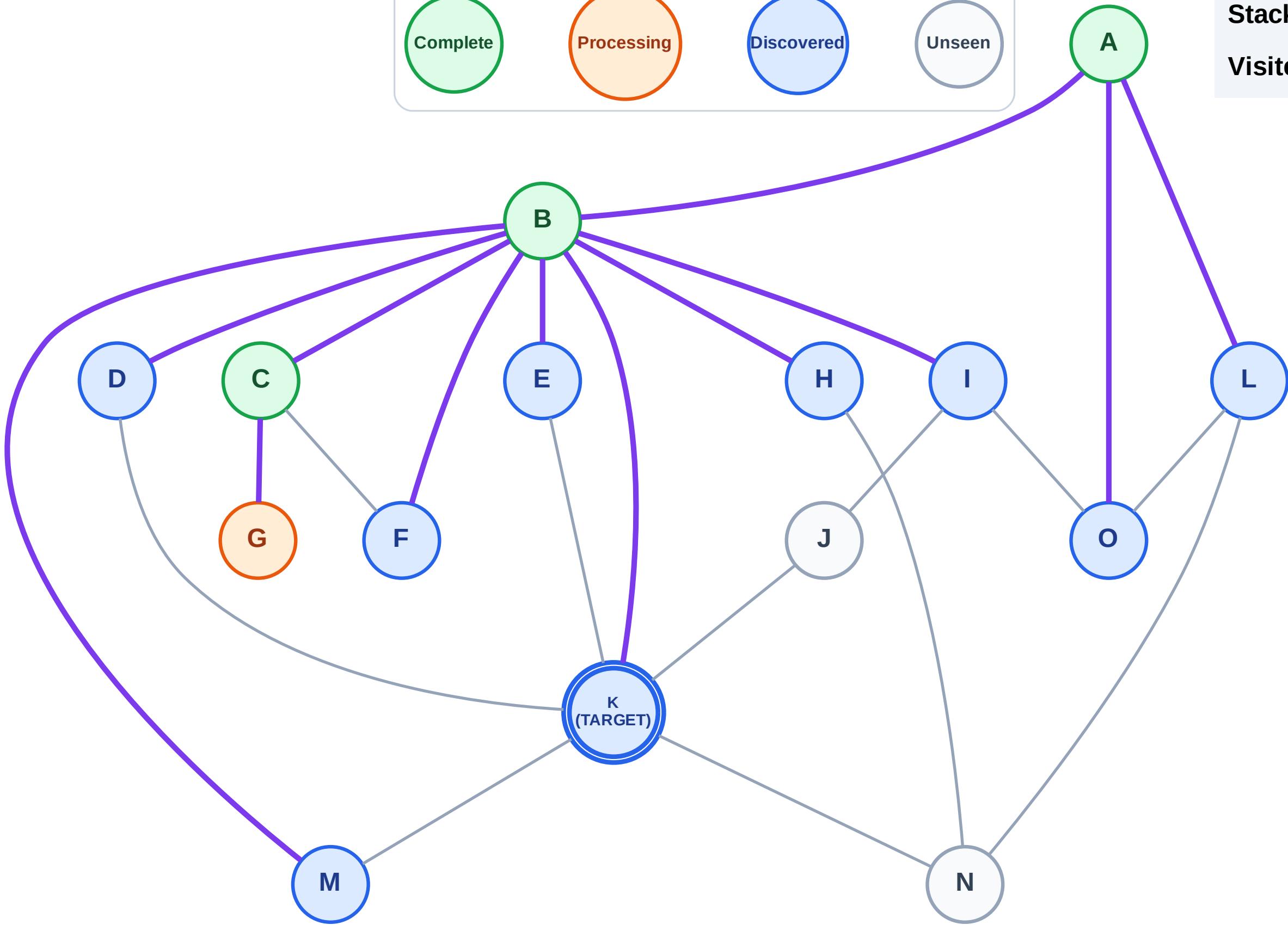
Processing

Discovered

Unseen

Stack (top at right): O | L | M | K | I | H | F | E | D

Visited: A, B, C



DFS Traversal

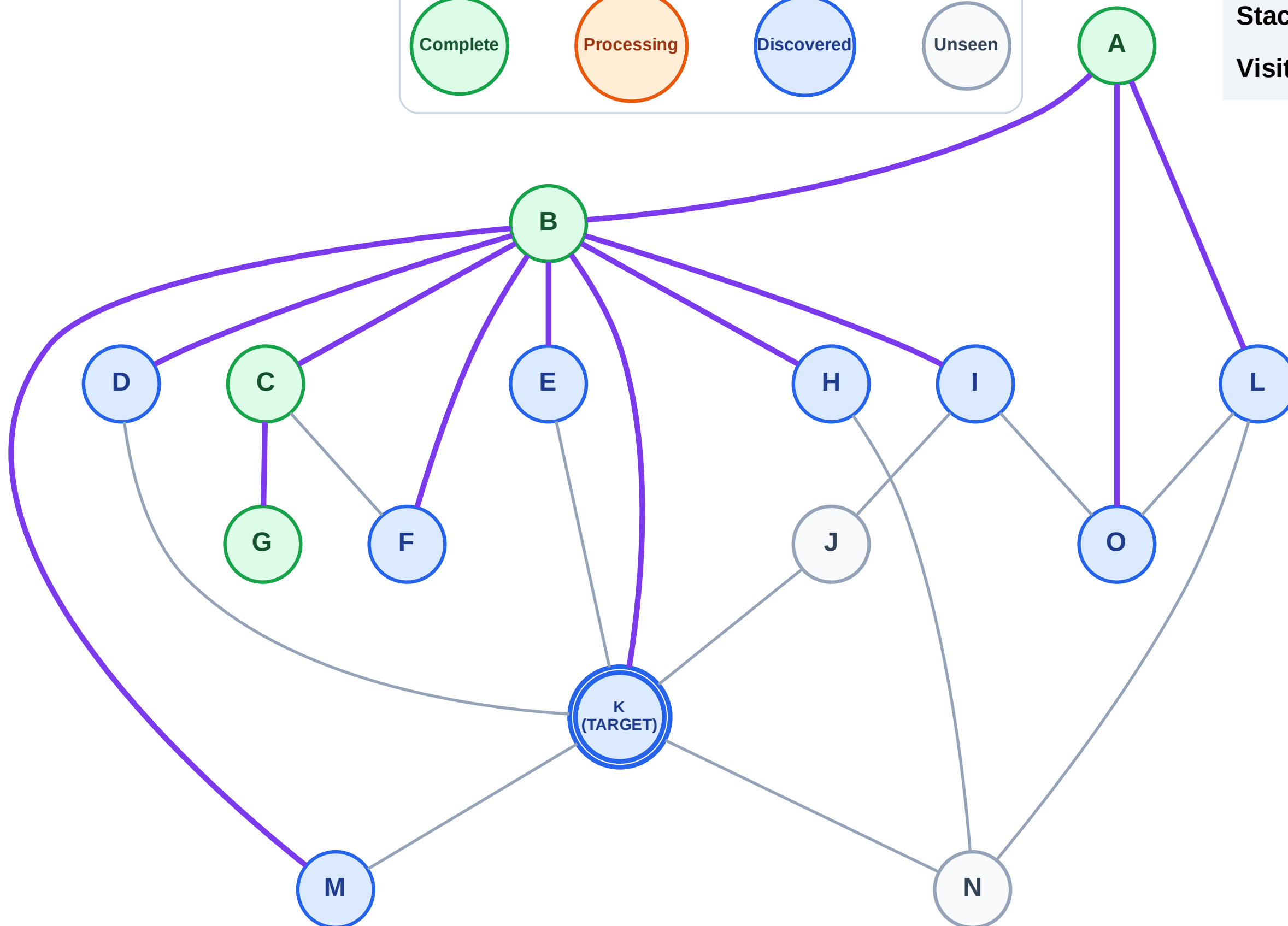
Step 9: No new neighbors from G

Legend



Stack (top at right): O | L | M | K | I | H | F | E | D

Visited: A, B, C, G



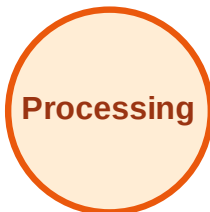
DFS Traversal

Step 10: Process node D

Legend



Complete



Processing



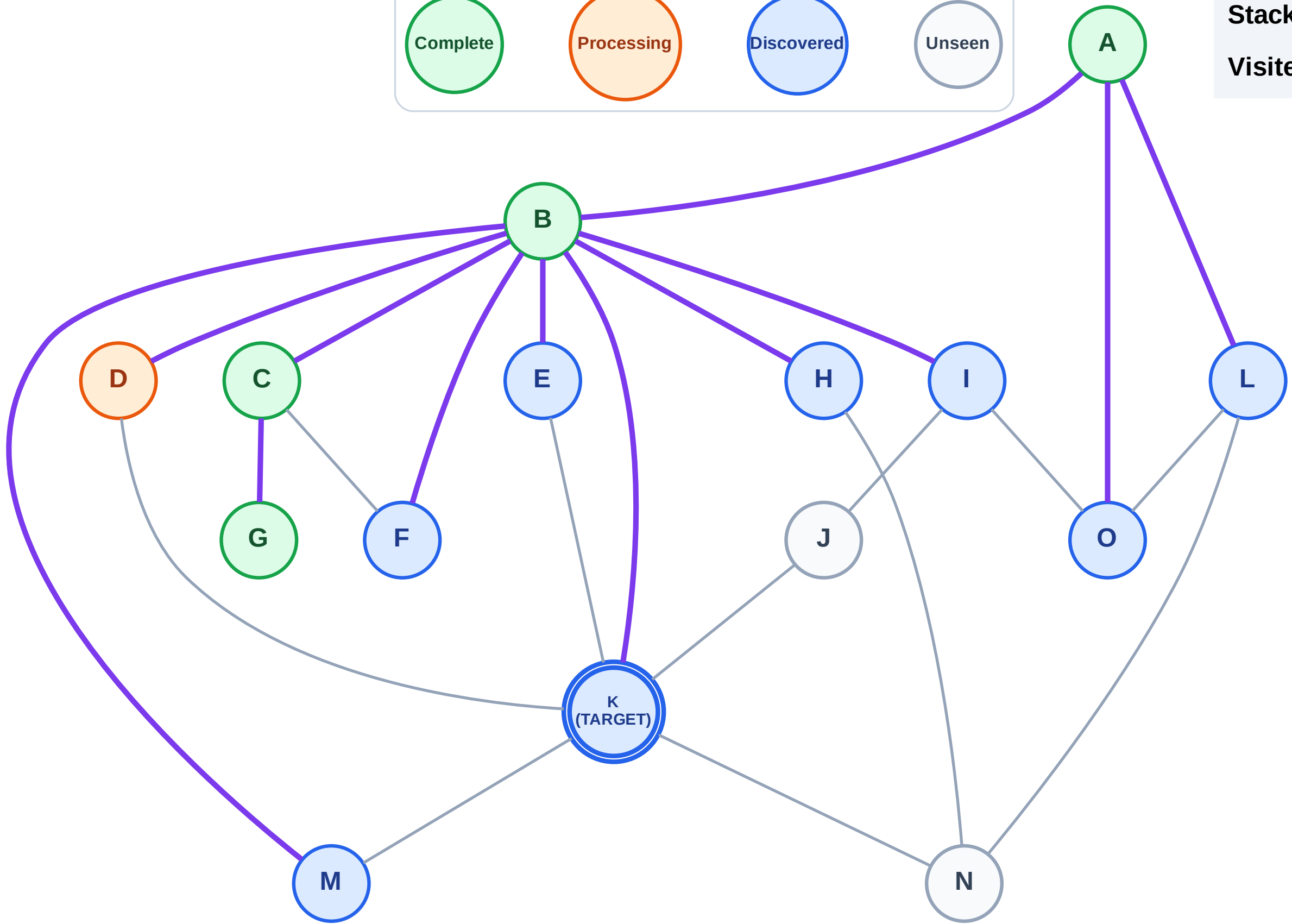
Discovered



Unseen

Stack (top at right): O | L | M | K | I | H | F | E

Visited: A, B, C, G



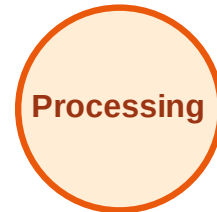
DFS Traversal

Step 11: No new neighbors from D

Legend



Complete



Processing



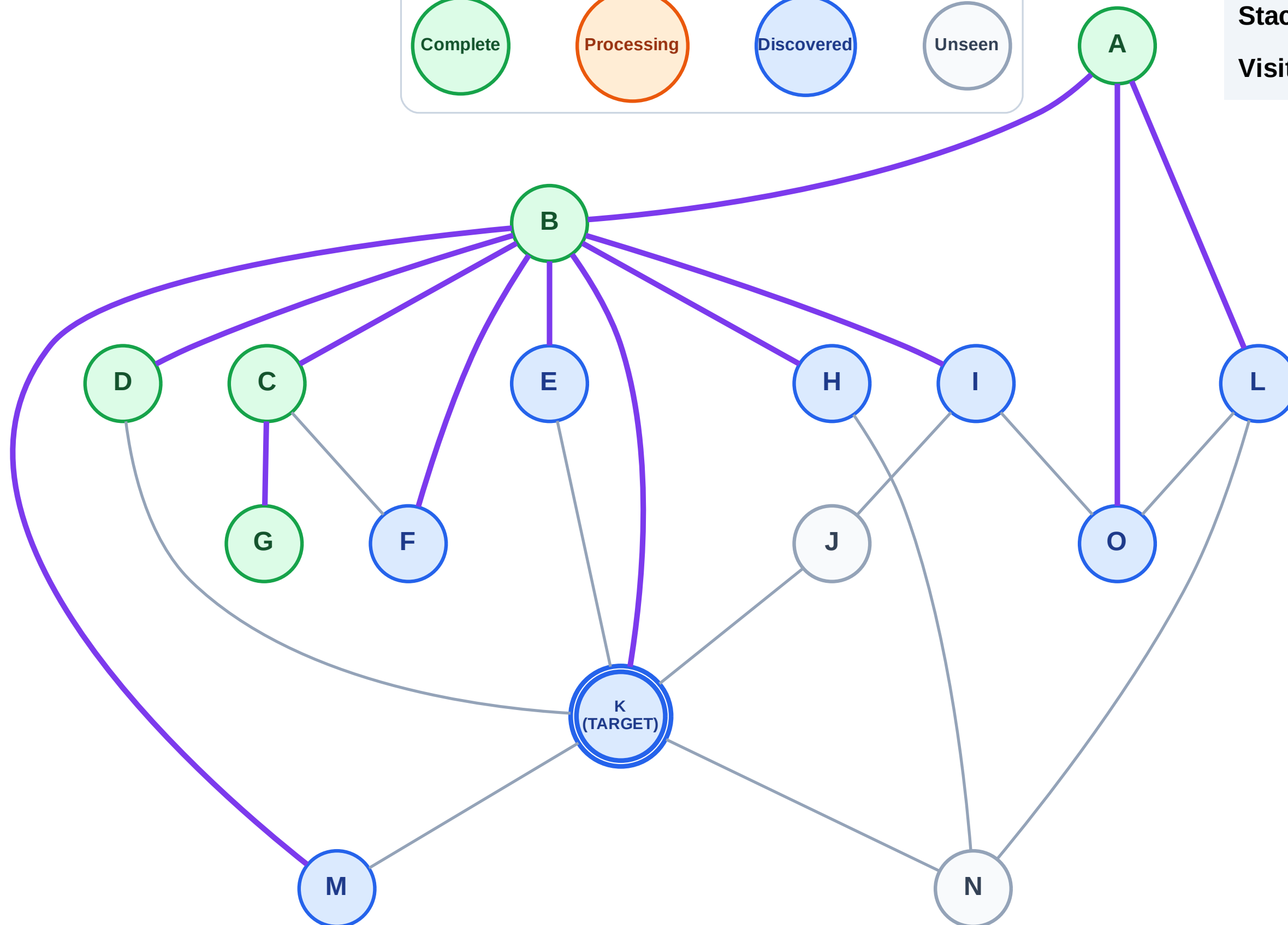
Discovered



Unseen

Stack (top at right): O | L | M | K | I | H | F | E

Visited: A, B, C, D, G



DFS Traversal

Step 12: Process node E

Legend



Complete



Processing



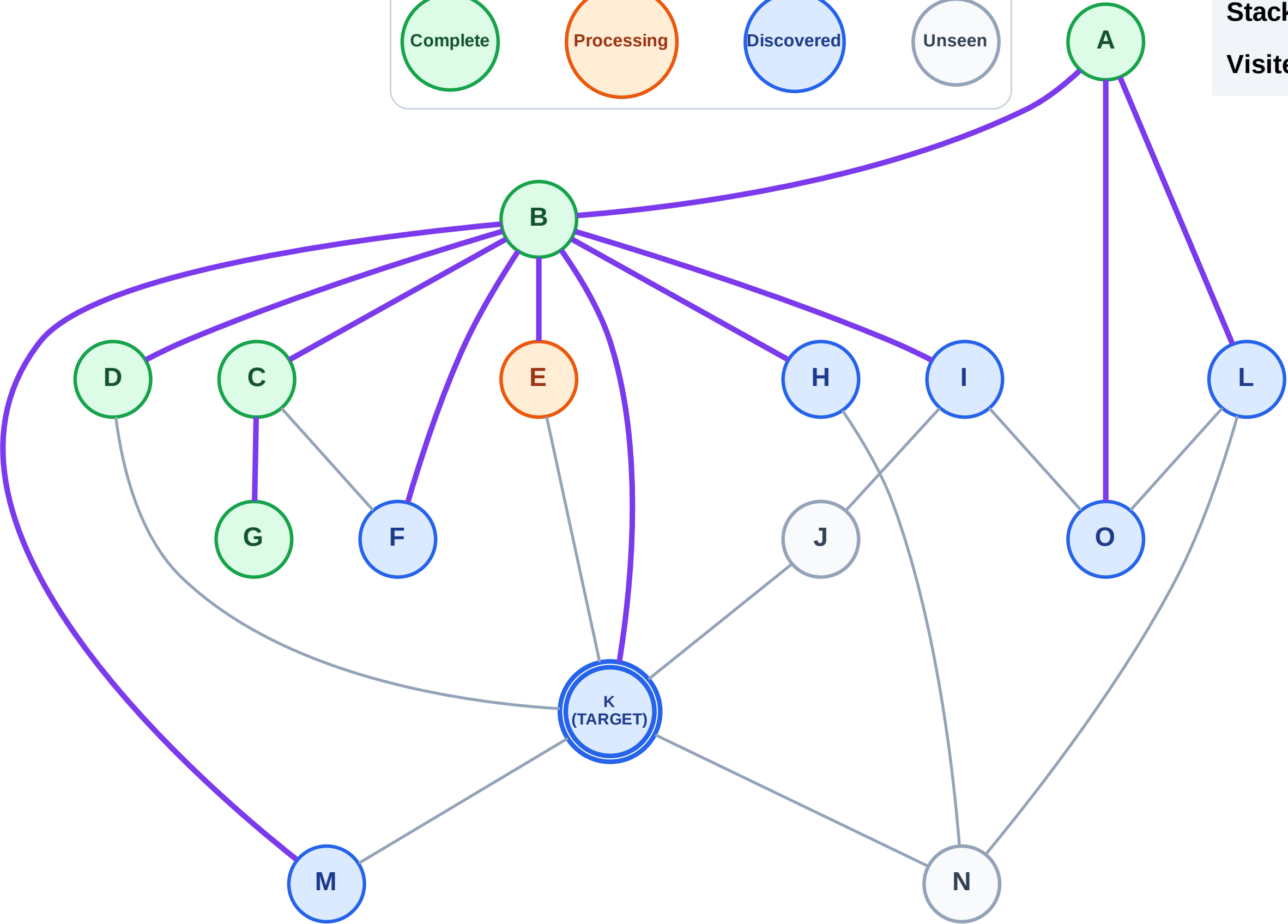
Discovered



Unseen

Stack (top at right): O | L | M | K | I | H | F

Visited: A, B, C, D, G



DFS Traversal

Step 13: No new neighbors from E

Legend



Complete



Processing



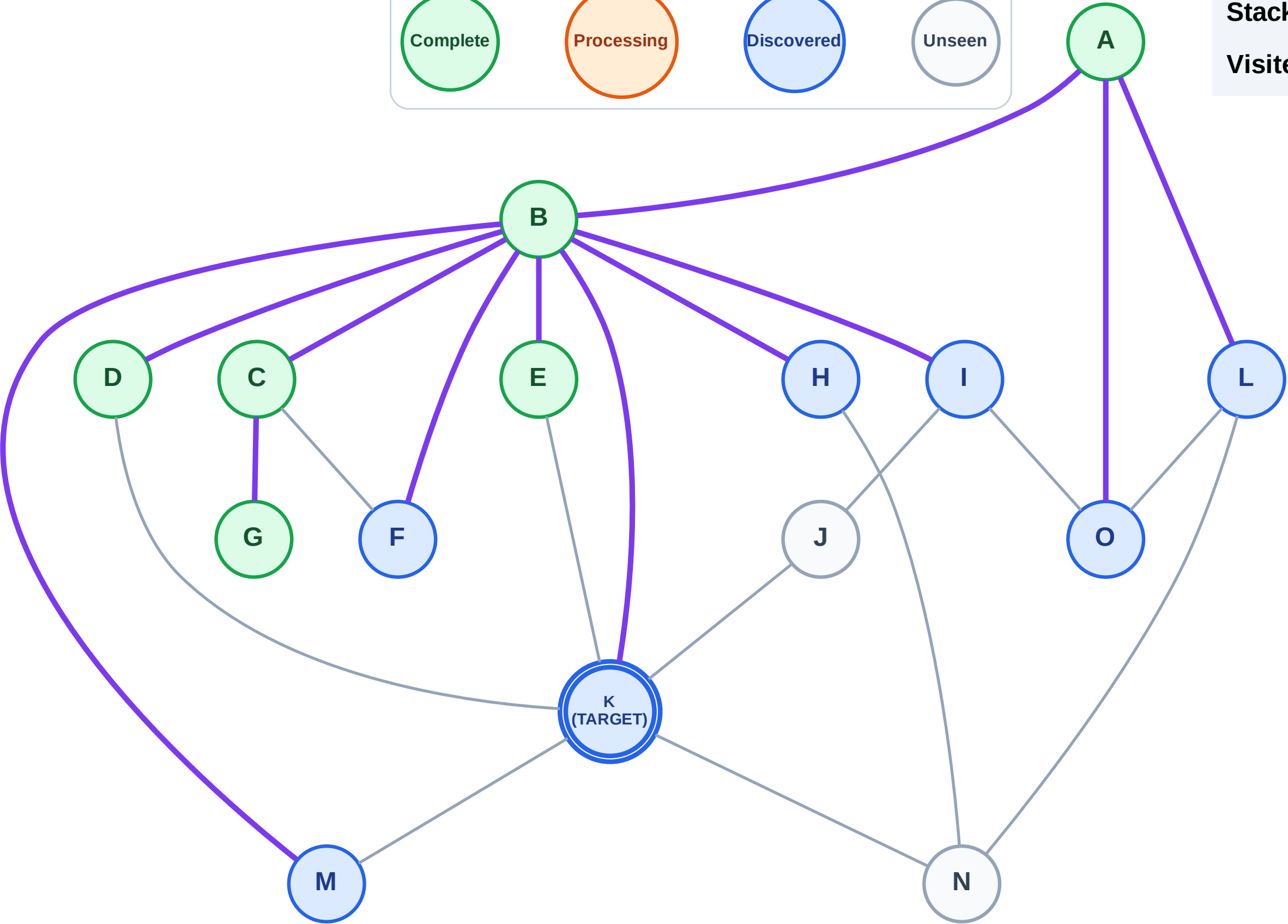
Discovered



Unseen

Stack (top at right): O | L | M | K | I | H | F

Visited: A, B, C, D, E, G



DFS Traversal

Step 14: Process node F

Legend

Complete

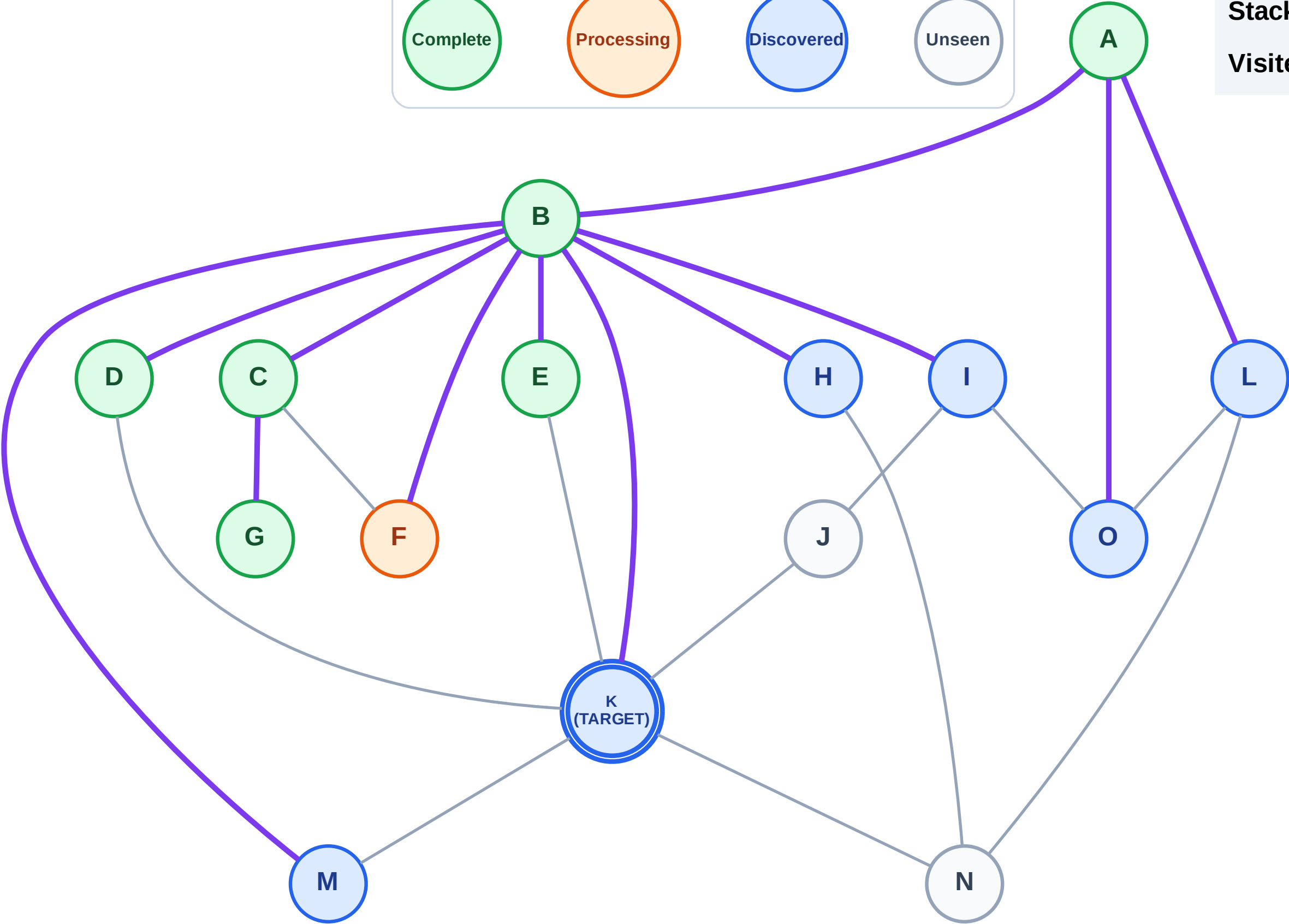
Processing

Discovered

Unseen

Stack (top at right): O | L | M | K | I | H

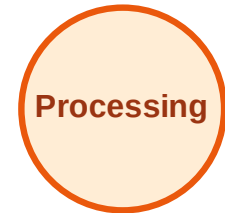
Visited: A, B, C, D, E, G



DFS Traversal

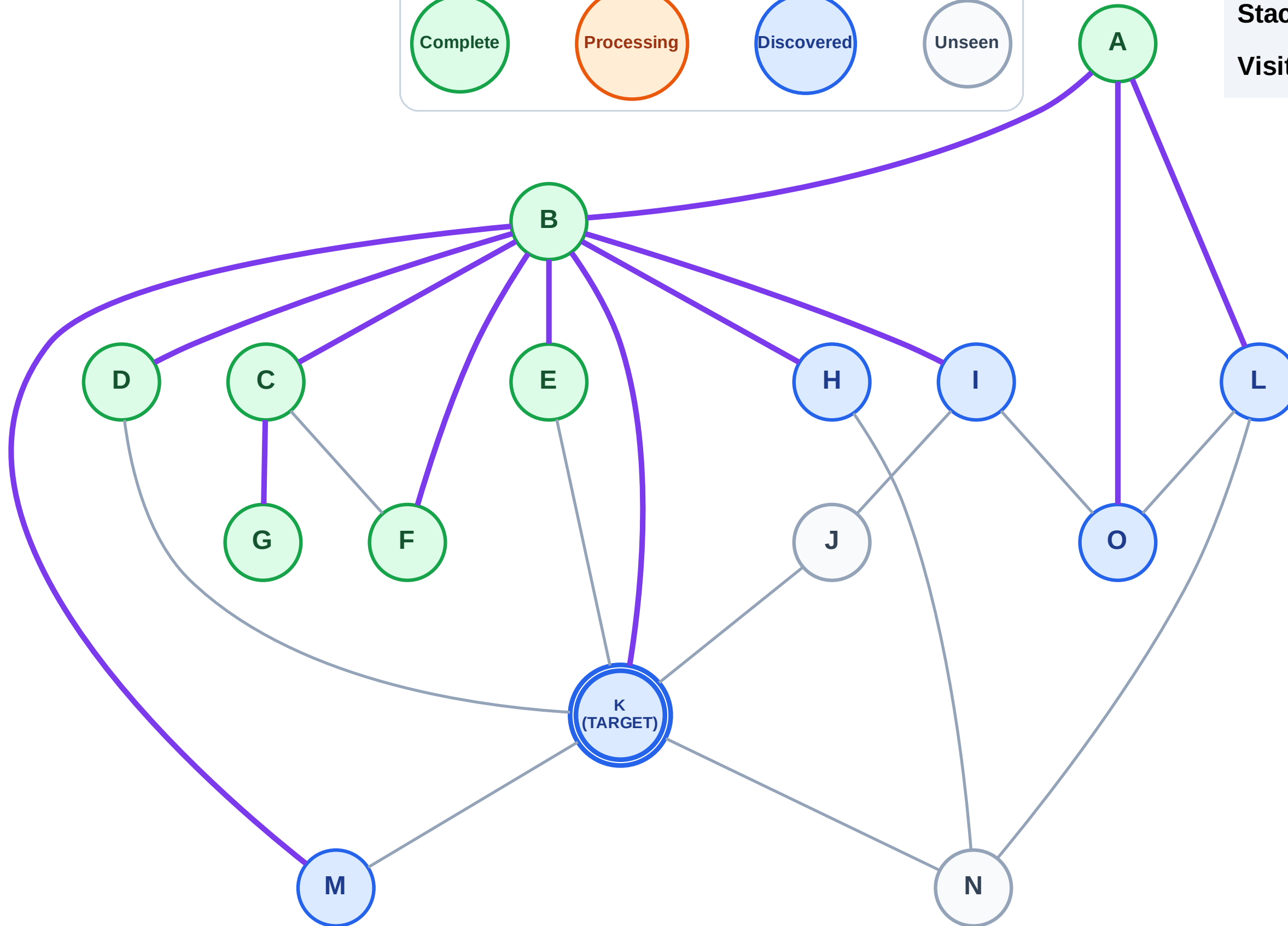
Step 15: No new neighbors from F

Legend



Stack (top at right): O | L | M | K | I | H

Visited: A, B, C, D, E, F, G



DFS Traversal

Step 16: Process node H

Legend

Complete

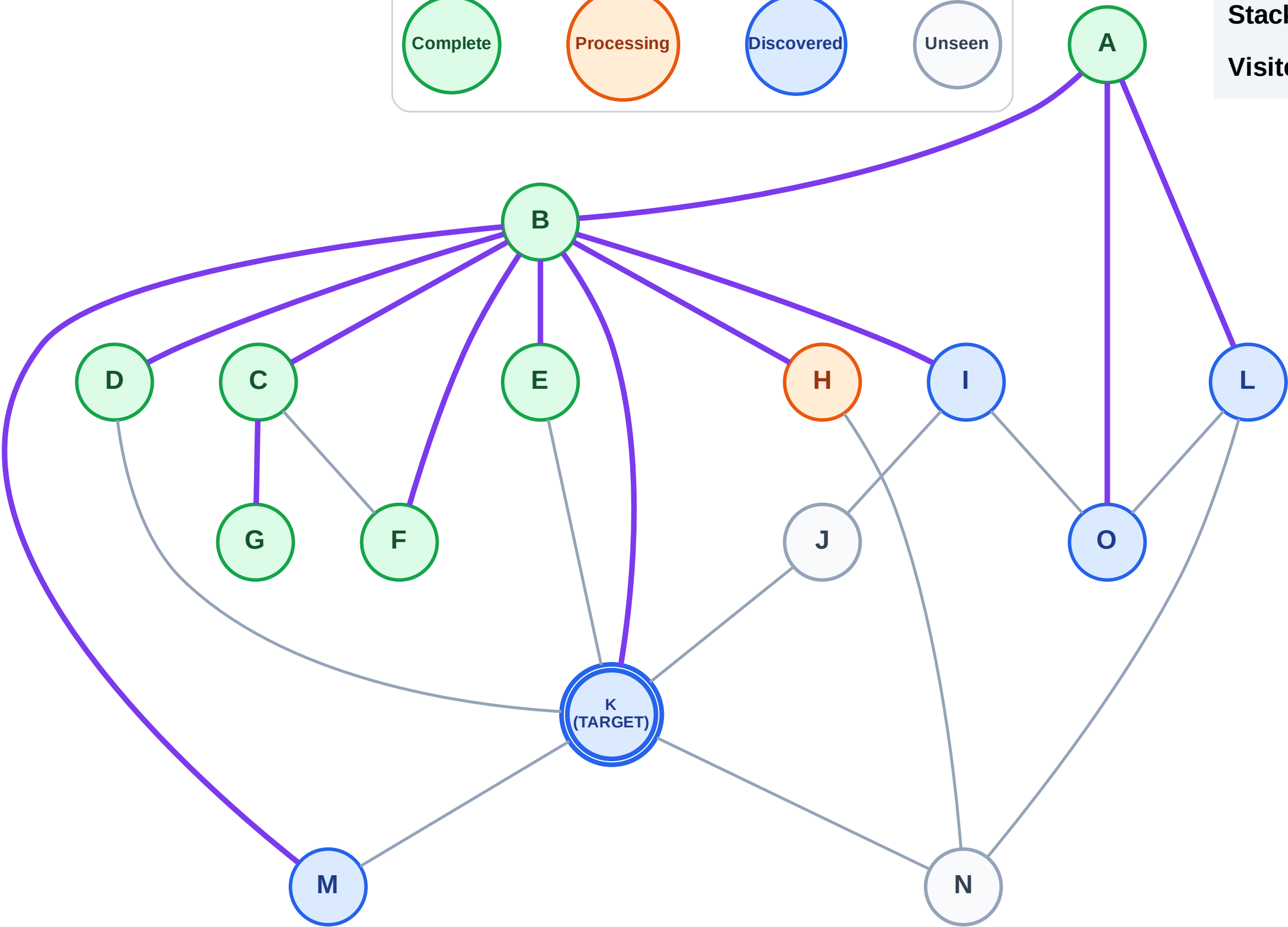
Processing

Discovered

Unseen

Stack (top at right): O | L | M | K | I

Visited: A, B, C, D, E, F, G



DFS Traversal

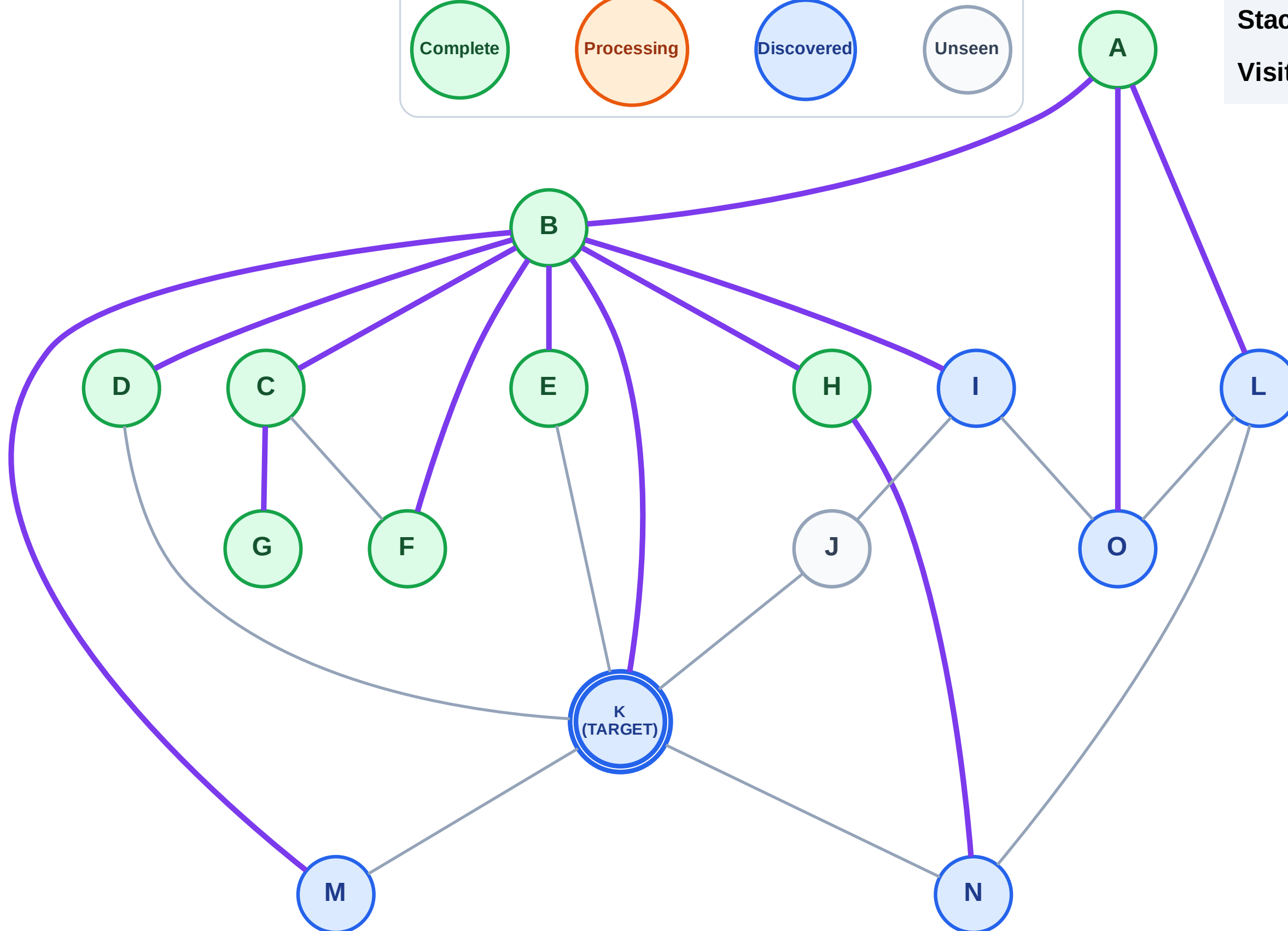
Step 17: Discover N from H

Legend



Stack (top at right): O | L | M | K | I | N

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node N

Legend

Complete

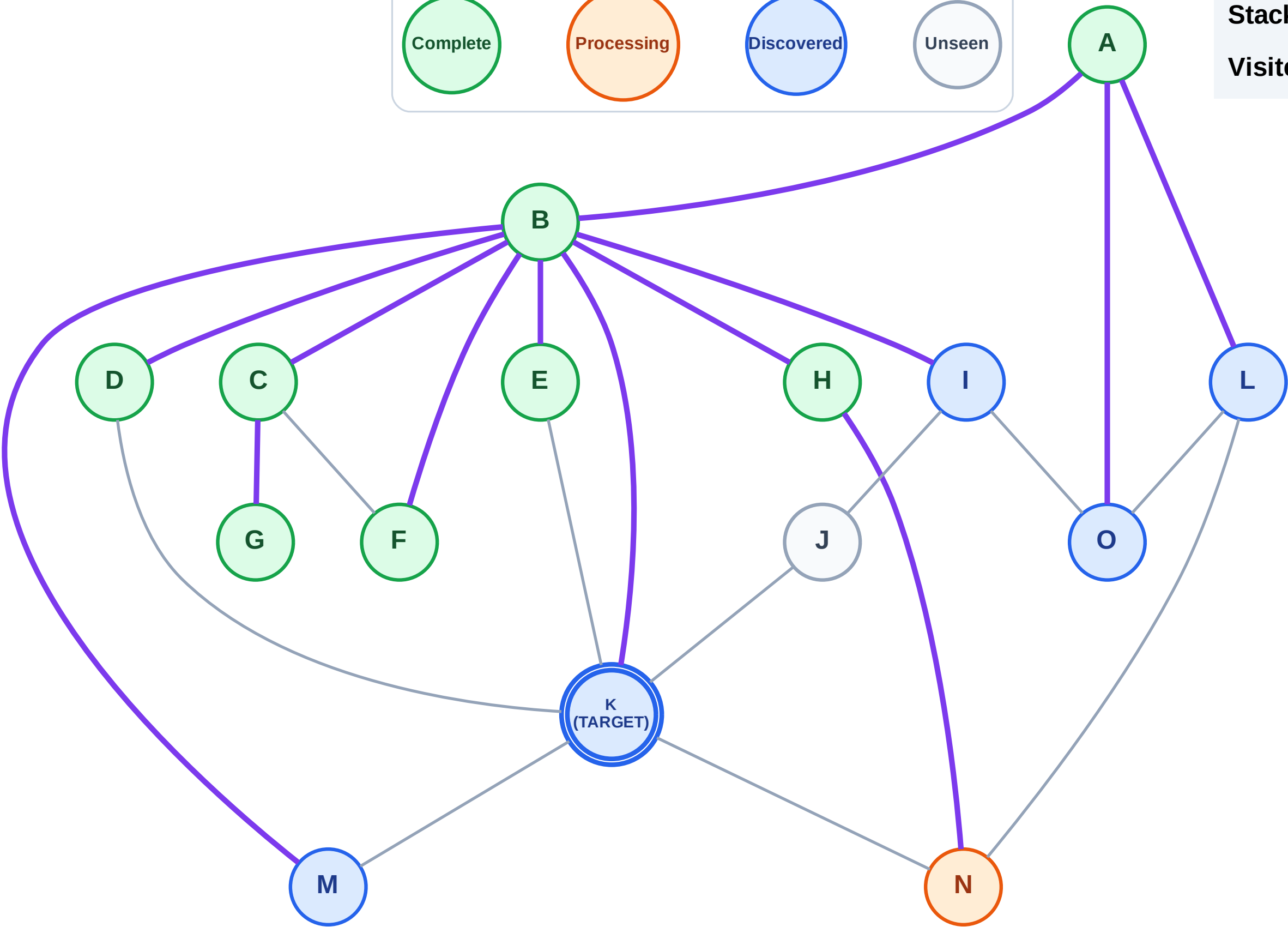
Processing

Discovered

Unseen

Stack (top at right): O | L | M | K | I

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 19: No new neighbors from N

Legend

Complete

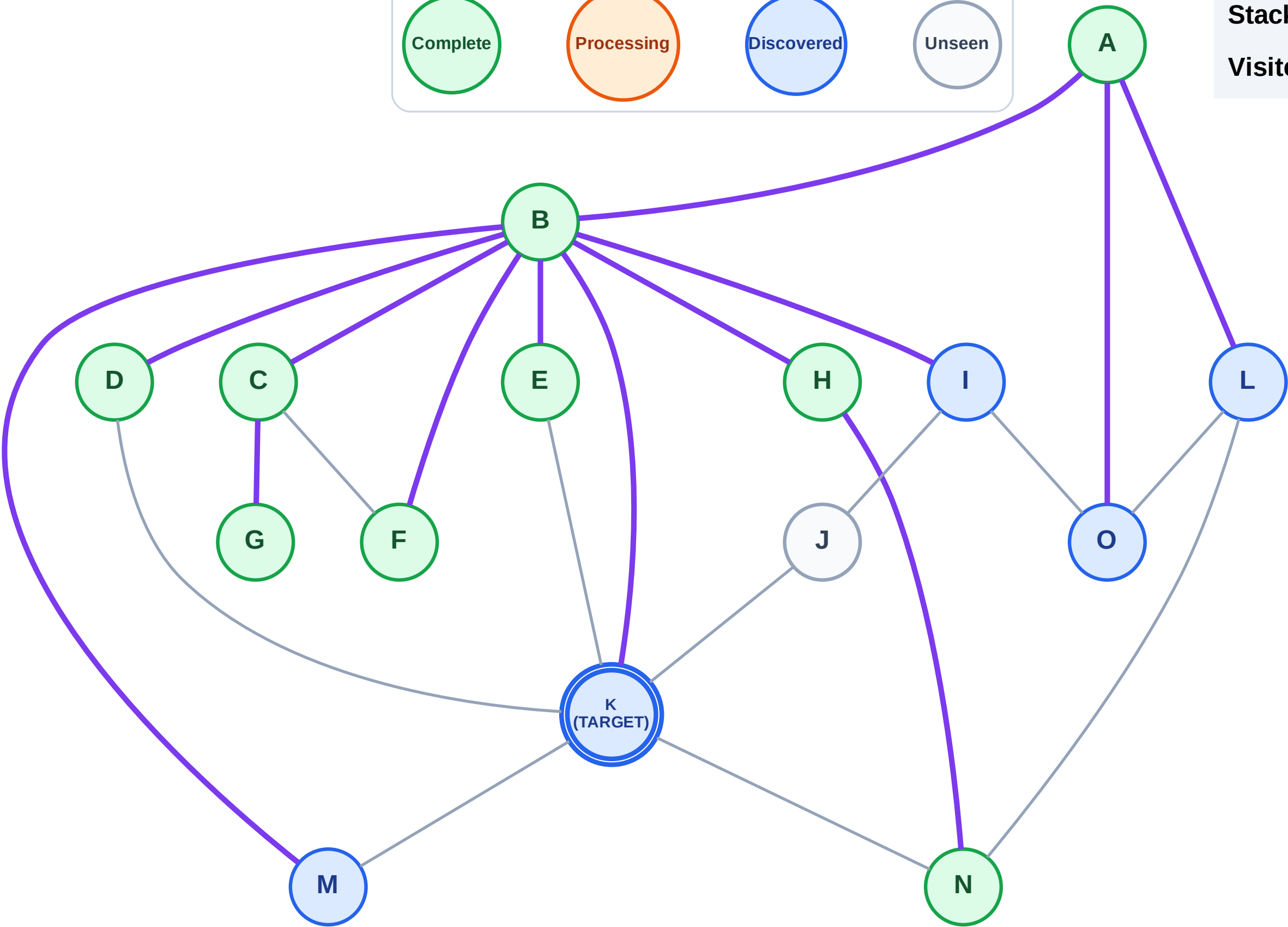
Processing

Discovered

Unseen

Stack (top at right): O | L | M | K | I

Visited: A, B, C, D, E, F, G, H, N

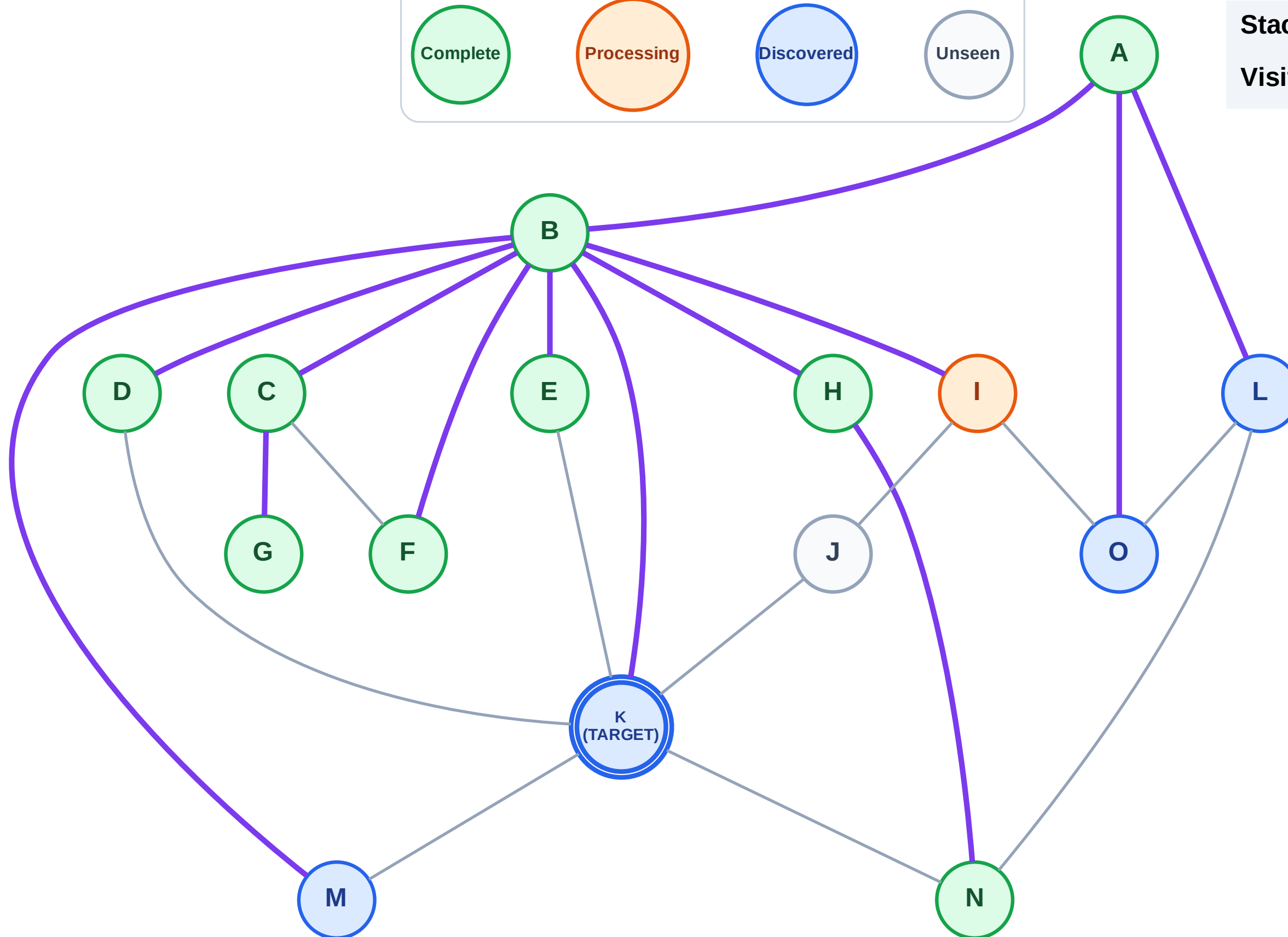


Step 20: Process node I

Legend

Unseen

Visited: A, B, C, D, E, F, G, H, N



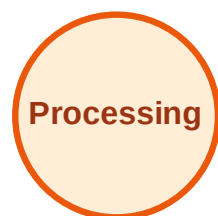
DFS Traversal

Step 21: Discover J from I

Legend



Complete



Processing



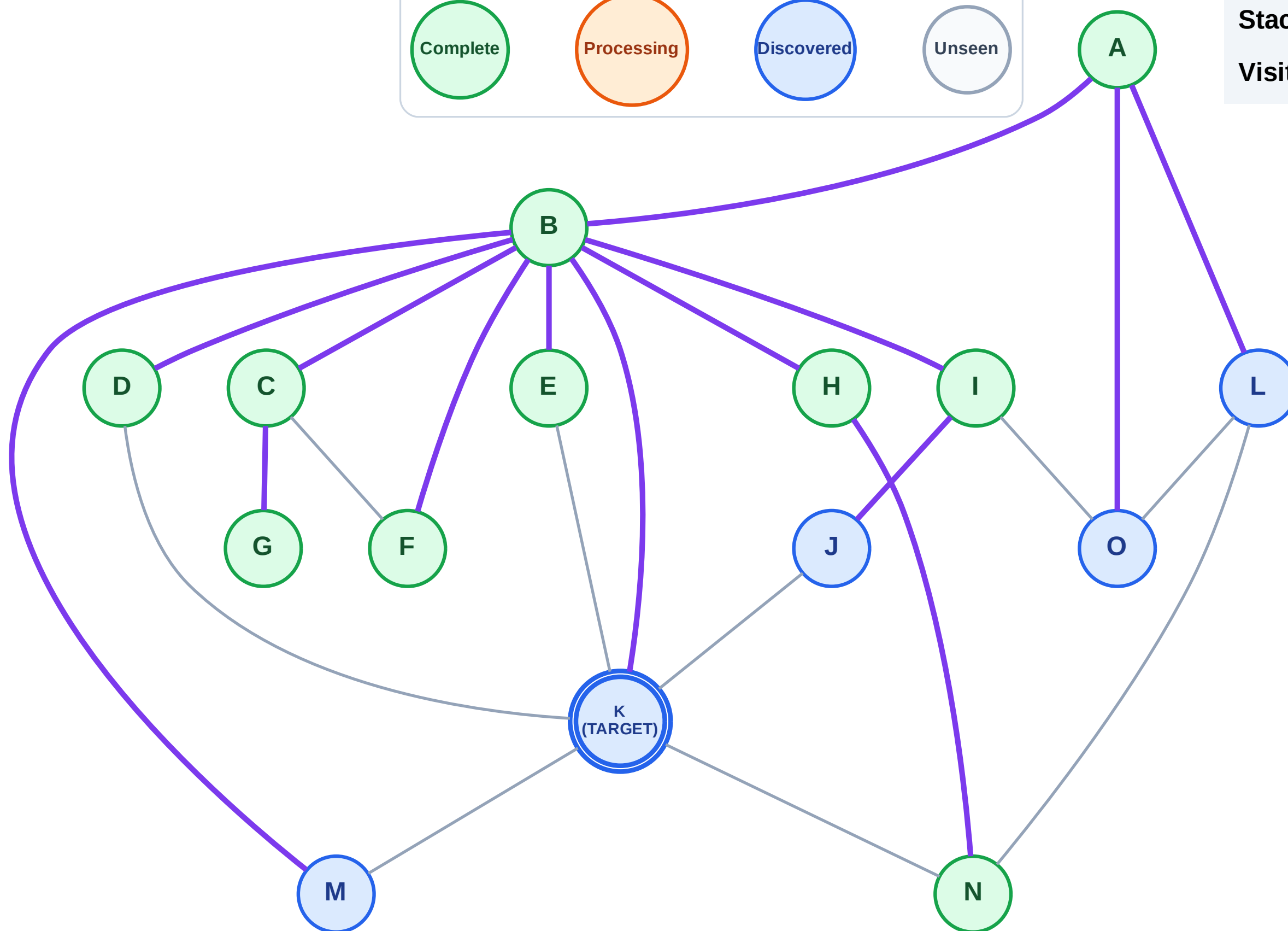
Discovered



Unseen

Stack (top at right): O | L | M | K | J

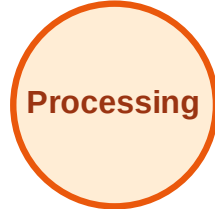
Visited: A, B, C, D, E, F, G, H, I, N



DFS Traversal

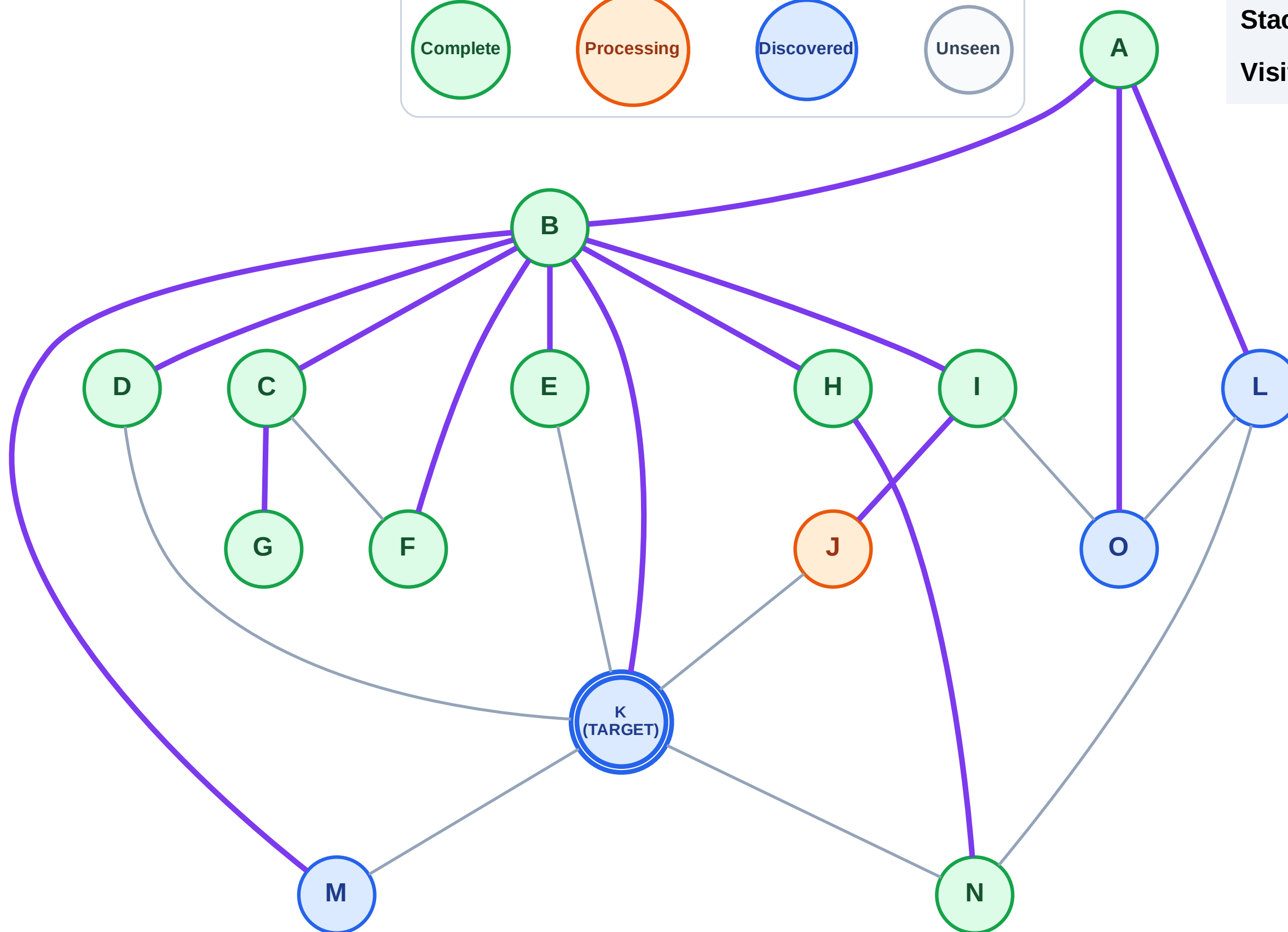
Step 22: Process node J

Legend



Stack (top at right): O | L | M | K

Visited: A, B, C, D, E, F, G, H, I, N



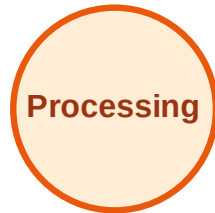
DFS Traversal

Step 23: No new neighbors from J

Legend



Complete



Processing



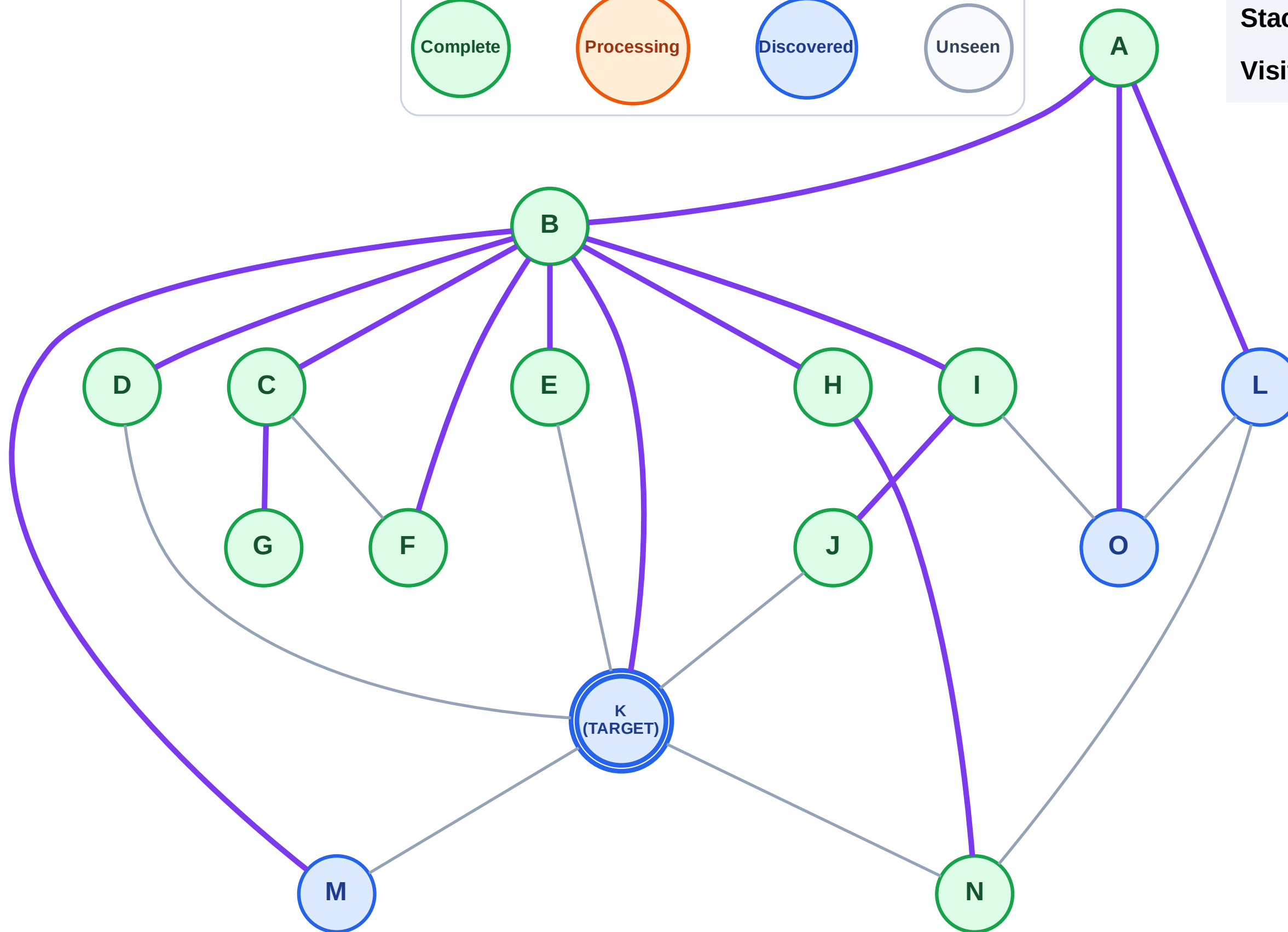
Discovered



Unseen

Stack (top at right): O | L | M | K

Visited: A, B, C, D, E, F, G, H, I, J, N



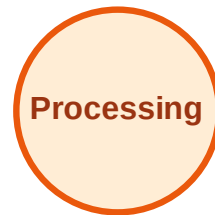
DFS Traversal

Step 24: Process node K

Legend



Complete



Processing



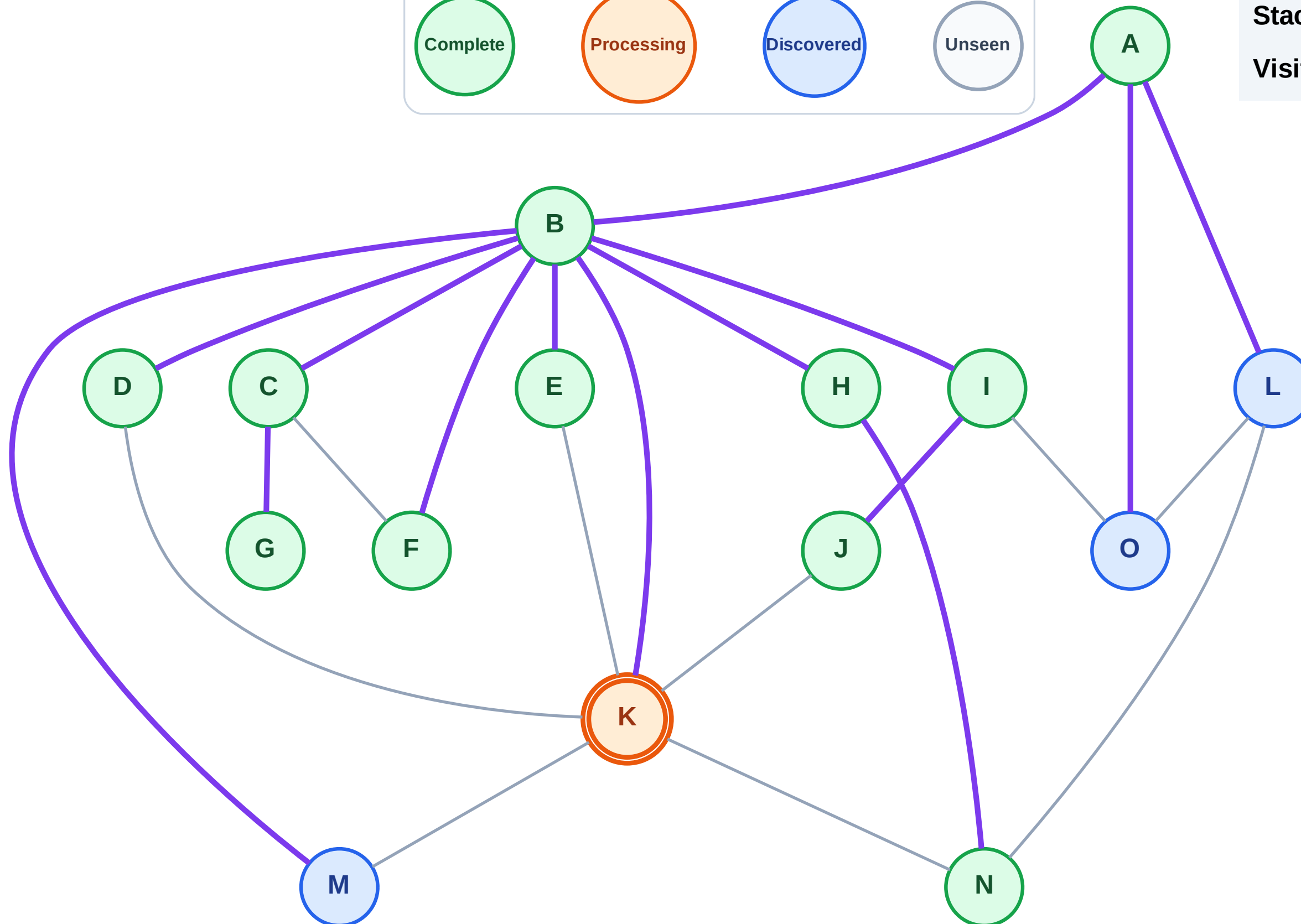
Discovered



Unseen

Stack (top at right): O | L | M

Visited: A, B, C, D, E, F, G, H, I, J, N



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | L | M

Visited: A, B, C, D, E, F, G, H, I, J, K, N

